A Deep Stochastic Adaptive Fourier Decomposition Network With Back-Propagation for Hyperspectral Image Classification

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***Abstract*—Convolutional neural networks (CNNs) have shown impressive performance in hyperspectral image (HSI) classification. However, these deep learning methods still face two major challenges. One is that they require a large number of training samples to train parameters, and the other is that high-dimensional nonlinear feature extraction and multisource information fusion. This article proposes a deep stochastic adaptive Fourier decomposition (SAFD) network integrated with back-propagation (BP) and multiscale feature fusion to significantly improve classification accuracy. The main contributions are threefold: 1) a deep SAFD network with BP is designed, which introduces BP into the deep SAFD network for the first time, and achieves automatic dynamic optimization of network parameters through the BP algorithm; 2) a Kalman filter-based multiscale pyramid construction method is proposed, which extracts hierarchical spatial features through state recursion equations and enhances texture representation by fusing local binary patterns (LBPs); and 3) an efficient classification algorithm based on a deep SAFD network with a BP algorithm is developed to integrate deep SAFD features, multiscale pyramid features, and LBP texture features, achieving higher classification accuracy. Experimental results show that the proposed method outperforms other selected HSI classification methods with similar principles. Moreover, compared with other state-of-the-art deep learning methods, the proposed method can achieve better classification performance.**

***Index Terms*—Deep learning, graph-based learning (GSL), hyperspectral image (HSI) classification, stochastic adaptive Fourier decomposition (SAFD).**

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I. Introduction

A. Research Background

With the advancement of remote sensing sensor technology, hyperspectral images (HSIs), which contain hundreds of contiguous spectral bands, have become increasingly accessible [1]. Unlike conventional RGB natural images, HSIs capture detailed spectral signatures across a wide range of wavelengths, providing a rich source of both spatial and spectral information [1], [2]. The primary goal of HSI classification is to assign each pixel in the image to one of several predefined land-cover categories [1], [3]. The abundance of spectral and spatial information offers more discriminative capacity for classification, making HSIs highly valuable in applications such as environmental monitoring, agriculture, and mineral exploration. Over the past two decades, numerous methods have been developed to address the challenges associated with HSI classification [4], [5], [6], [7].

B. Problem Definition and Challenges

The core challenge in HSI classification stems from the inherent high dimensionality and complexity of the data. While the abundant spectral and spatial information provides great discriminative potential, it also introduces significant difficulties. Specifically, three critical challenges persist as follows.

1) Data Hunger of Deep Models: Deep learning approaches typically require a large number of labeled training samples to optimize their numerous parameters, which are often scarce and expensive to obtain in remote sensing applications.

2) Limited Adaptability of Pretraining Strategies: Existing parameter pretraining methods often rely on manually preset hyperparameters that lack adaptability to diverse data characteristics, limiting their feature representation capacity.

3) Complex Feature Extraction and Fusion: Effectively extracting high-dimensional nonlinear features and fusing multisource information (spectral, spatial, and textual) remains challenging for current methods.

These challenges are particularly evident when dealing with complex HSI scenarios, where traditional feature engineering approaches suffer from performance degradation due to limited generalization capability and insufficient deep feature representation.

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TABLE I
SYSTEMATIC COMPARISON OF MAINSTREAM HSI CLASSIFICATION METHODS

Category	Core Principle	Advantages	Disadvantages
Traditional machine learning (e.g., SVM, RF)	Handcrafted features and shallow classifier.	Good performance with small samples; Interpretable; Low computational cost.	Relies heavily on expert knowledge for feature engineering; Limited ability to capture deep, hierarchical features; Poor generalization to complex scenes.
CNN-based Methods	Hierarchical local feature extraction via convolutional kernels.	Powerful automatic spatial-spectral feature learning; Strong performance on large datasets.	Limited receptive field struggles with global dependencies; High computational cost for 3D convolutions; Prone to overfitting with limited labeled data.
Transformer-based Methods	Global context modeling via self-attention mechanisms.	Excellent at capturing long-range dependencies; Superior global representation ability.	Extremely data-hungry; High computational complexity; Lacks innate spatial inductive bias; Gradient vanishing in deep layers.
Pre-training Strategies	Initialize parameters using unsupervised or self-supervised rules.	Reduces reliance on labeled data; Accelerates training convergence.	Manual parameter setting lacks adaptability; Pre-training objectives may misalign with downstream tasks; Limited feature representation capacity.

C. Systematic Review and Limitations of Existing Works

To provide a clearer context for our work, we first present a systematic comparison of mainstream HSI classification paradigms in Table I, summarizing their core principles, advantages, and disadvantages. This is followed by a detailed discussion of their specific limitations.

1) Traditional Methods: Early approaches (e.g., support vector machine (SVM) [8], random forest (RF) [9]) typically involved two sequential stages: feature extraction and classifier training. Gao et al. [10] proposed a spectral and spatial classification framework using random multigraphs. However, as summarized in Table I, these methods are limited as they primarily extract low-level features and depend on handcrafted features that lack robustness in complex situations [5]. Object-level spatial feature aggregation approaches such as principal component analysis (PCA) [11], local binary pattern (LBP) [12], and Kalman filtering [13] demonstrated improvements but still suffered from limited generalization and insufficient deep feature representation.

2) Deep Learning Approaches: Convolutional neural networks (CNNs) have been widely adopted for HSI classification [4], [5], [14], [15], [16] due to their ability to extract hierarchical local features. Various architectures have been explored, including 1-D-CNNs [17], 2D-CNNs [18], and 3D-CNNs [19], [20] that extract joint spectral–spatial features. Liu et al. [21] developed a multiscale large kernel CNN (MSLKCNN) for HSI classification, which leverages large kernel convolutions (up to 15×15) to capture both local and global spectral–spatial features without the need for complex architectural designs. Hybrid approaches combining 2-D and 3-D CNNs [22], [23], and attention-enhanced CNNs [23] have also been proposed.

Recent advances in CNN-based HSI classification have focused on integrating CNNs with other architectures to overcome their inherent limitations. For instance, Cui et al. [24] proposed a CNN-enhanced hypergraph attention network (CEHGAT) that combines a hypergraph attention branch with a multiscale CNN branch to capture both high-order interactions and spatial–spectral features. Similarly, Shi et al. [25] introduced a hybrid CNN-GCN network (HCGN) that leverages graph convolutional networks to enhance superpixel-level features while using CNNs for pixel-level fine feature extraction. Another notable approach is the Mamba-CNN framework proposed by Du et al. [26], which integrates Mamba state space models with CNNs to efficiently capture long-range dependencies while maintaining local feature extraction capabilities. Furthermore, Li et al. [27] developed a CNN and dual Swin-transformer fusion network (CSTFNet) that combines the strengths of CNNs in local feature extraction with the global modeling capabilities of transformers.

While CNNs excel at local feature extraction, they struggle to capture global representations and long-range dependencies, as indicated in Table I. Enlarging the receptive field increases computational cost significantly [28], [29], [30], [31].

3) Transformer-Based Methods: Recently, transformers have been introduced to computer vision tasks, including HSI classification [1], [32], [33], [34] due to their capability of capturing long-range dependencies. Vision transformer (ViT) [35] inspired several HSI-specific adaptations [1], [32], [33], [34], [36], [37]. Despite their remarkable progress, these methods face challenges such as gradient vanishing, limited interpretability, and substantial sample requirements, as systematically compared in Table I. Moreover, they often adopt simple concatenation strategies that fail to fully exploit complementary relationships among multiscale features.

4) Pretraining Strategies: To reduce the reliance on extensive labeled samples, several parameter pretraining strategies have been proposed [5], [38], [39], [40]. Based on stochastic adaptive Fourier decomposition (SAFD) [41], [42], [43], [44], [45], Cheng et al. [46] proposed a deep SAFD network that optimizes parameters through SAFD. While these methods reduce training time, they still require laborious manual parameter adjustment and lack adaptability to diverse data characteristics, which is a key limitation we aim to address.

D. Our Contributions and Proposed Solution

To address the aforementioned limitations systematically summarized in Table I, we propose a novel deep SAFD with back-propagation (BP) network that automatically adjusts parameters in an unsupervised manner using the SAFD with BP algorithm. This approach eliminates manual parameter tuning while reducing the need for large labeled datasets. Our

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solution comprises three key innovations, each designed to overcome specific drawbacks of existing methods:

First, to tackle the limitations of fixed pretraining and enhance feature extraction, we generalize the 1-D Kalman filter to its 3-D counterpart to construct a multiscale pyramid for HSIs, enabling effective acquisition of high-dimensional nonlinear features. Second, to enhance robustness to rotational variations—a challenge not fully addressed by existing CNN or transformer methods—we incorporate LBP processing to derive rotation-invariant texture features. Finally, we develop an efficient classification framework that integrates deep features learned by our adaptive SAFD-BP network, multiscale Kalman pyramid features, and rotation-invariant LBP descriptors through a sophisticated fusion strategy that fully exploits complementary information, moving beyond simple concatenation.

The main contributions and novelty of this article are summarized as follows.

1) *Addressing Manual Parameter Limitations:* Current SAFD networks suffer from reliance on empirically preset parameters that lack adaptability to diverse data characteristics. To overcome this critical limitation, we introduce a deep SAFD network integrated with BP. This novel architecture enables automatic, dynamic optimization of network parameters through BP learning, eliminating the need for manual tuning and significantly enhancing model adaptability and representation power.
2) *Solving High-Dimensional Pyramid Construction Challenges:* Traditional pyramid methods struggle to effectively capture spatial–spectral features in HSIs due to their high dimensionality and complex local structures. We propose a multiscale pyramid construction method based on a Kalman filter that iteratively predicts detail layers using state equations. Crucially, we extend the 1-D Kalman filter to 3-D, enabling robust multiscale decomposition specifically for HSIs. Furthermore, to enhance robustness against rotational variations in textures, we incorporate LBP processing, deriving rotation-invariant LBP features from the pyramid levels.
3) *Integrating Disparate Feature Representations:* Existing HSI classification methods often fail to fully exploit complementary information from deep features, multiscale contextual features, and invariant texture descriptors. We construct an efficient classification algorithm based on our deep SAFD network with BP. This framework integrates learned deep features, multiscale Kalman pyramid features, and rotation-invariant LBP features into a unified model. This synergistic fusion directly addresses the feature representation gap, leading to significantly enhanced classification performance.

The remainder of this article is organized as follows. Section II introduces the preliminary knowledge of SAFD. Section III-A presents the proposed SAFD model with BP for HSI, which is designed to learn deep features from HSIs. Sections III-B, III-D, and III-E present Kalman filter-based multiscale pyramid construction for HSIs, spatial–spectral feature extraction, complexity analysis of feature extraction, and the classification framework of HSIs, respectively. Sections IV and V are experiments and conclusion, respectively.

II. Relevant Background

As the proposed deep network framework in Section III relies on SAFD, this section first elaborates the SAFD theoretical foundations, which has appeared in literature [41], [42], [45], [47].

For a group of analytic random signals (RSs) $f^i(i = 1, 2, \ldots, N)$ in Hardy space $H^2(\mathbb{D})$ and the unit disk $\mathbb{D}$ in the complex plane. The first step of the SAFD is

$$\begin{aligned} f^i(z) = f_1^i(z) &= \left(f_1^i(z) - \left\langle f_1^i, e_{a_1}\right\rangle e_{a_1}(z)\right) + \left\langle f_1^i, e_{a_1}\right\rangle e_{a_1}(z) \\ &= f_2^i(z)\frac{z - a_1}{1 - \bar{a}_1 z} + \left\langle f_1^i, e_{a_1}\right\rangle \mathcal{B}_{a_1}(z) \end{aligned} \tag{1}$$

where

$$f_2^i(z) = \frac{f_1^i(z) - \left\langle f_1^i, e_{a_1}\right\rangle e_{a_1}(z)}{\frac{z-a_1}{1-\bar{a}_1 z}} \quad \forall a_1 \in \mathbb{D}. \tag{2}$$

Applying (1) iteratively n times yields the following expression:

$$f^i(z) = f_{n+1}^i(z)\prod_{k=1}^{n}\frac{z - a_k}{1 - \bar{a}_k z} + \sum_{k=1}^{n}\left\langle f_k^i, e_{a_k}\right\rangle \mathcal{B}_k(z) \tag{3}$$

where

$$\mathcal{B}_k(z) = e_{a_k}(z)\prod_{l=1}^{k-1}\frac{z - a_l}{1 - \bar{a}_l z} \tag{4}$$

and

$$f_k^i(z) = \frac{\left(f_{k-1}^i(z) - \left\langle f_{k-1}^i, e_{a_{k-1}}\right\rangle e_{a_{k-1}}(z)\right)}{\frac{z-a_k}{1-\bar{a}_k z}} \in H^2(\mathbb{D}). \tag{5}$$

The set of analytic and rational functions $\{\mathcal{B}_k\}_{k=1}^{\infty}$ is the automatically generated orthonormal Takenaka–Malmquist (TM) system corresponding to the maximally selected a_k $(k = 1, 2, 3, \ldots) \in \mathbb{D}$. To obtain the maximum energy, the stochastic maximal selection principle (SMSP) is adopted to select the optimal parameter

$$a_k = \arg\max\left\{E_i\left|\left\langle f_k^i, e_a\right\rangle\right|^2, a \in \mathbb{D}\right\}. \tag{6}$$

Here, the expectation $E_i|\cdot|^2$ is introduced, indicating that

$$E_i\left|\left\langle f_k^i, e_a\right\rangle\right|^2 = \frac{1}{N}\sum_{i=1}^{N}\left|\left\langle f_k^i, e_a\right\rangle\right|^2. \tag{7}$$

Literature [47] provided the existence of a_k and the convergence proof of the formula.

III. Proposed SAFD Network With BP Mechanism for HSI Classification

In our previous research, we established the SAFD network [46] that employs convolutional operations and rectified linear unit (ReLU) activation functions as fundamental modules for deep feature extraction without using BP algorithms. The absence of the BP algorithm in the original SAFD framework, where initial network weights were trained without subsequent fine-tuning, leads to inherent limitations in feature discriminability. Specifically, this architectural constraint necessitates

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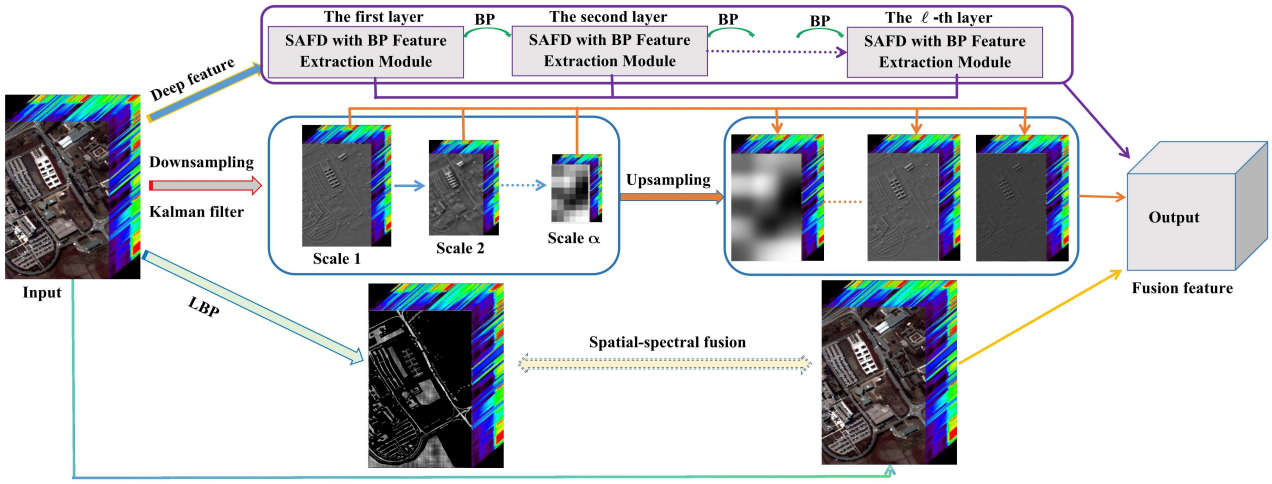


Fig. 1. Schematic of the proposed SAFD network with BP structure for feature extraction of HSI, and the detailed architecture of the SAFD with BP feature extraction module is shown in Fig. 2. The feature extraction process of the proposed method comprises three components: deep feature extraction using the SAFD network with BP algorithm; multiscale pyramid construction through Kalman filtering; spatial–spectral feature generation by integrating LBP-based spatial features with spectral characteristics. These three distinct feature modalities are subsequently fused to form a novel multisource composite feature representation.

the employment of significantly larger convolutional kernels and a greater number of feature channels to achieve comparable classification performance to that of deep learning methods. In contrast, deep networks incorporating BP mechanisms demonstrate superior discriminative power through their capacity for adaptive feature refinement. The gradient-based optimization enables hierarchical feature representations that are more aligned with semantic distinctions in the data manifold, as evidenced by comparative studies on feature separability indices.

To address these limitations, this section presents a novel enhanced SAFD with a BP algorithm. Distinct from this existing architecture, the proposed method systematically integrates BP mechanisms while maintaining unsupervised learning characteristics. This novel architecture implements end-to-end hierarchical feature learning where each subsequent layer performs parameter optimization on features from the preceding layer through gradient-based propagation. Notably, the BP process operates without utilizing labeled samples, thereby preserving the network's unsupervised feature learning paradigm. The proposed framework achieves layer-wise feature refinement through this self-supervised backward propagation mechanism while maintaining compatibility with unsupervised learning constraints.

Furthermore, to enhance the proposed network's discriminative capacity, we innovatively integrate multiscale information fusion into the deep feature learning pipeline. This is achieved through a Kalman filtering-based hyperspectral feature pyramid construction mechanism, which establishes interscale correlations while preserving spectral–spatial consistency. The pyramid architecture generates a multiscale feature hierarchy spanning from localized texture patterns to global contextual representations.

Building upon this multimodal feature foundation, we subsequently develop a semi-supervised graph learning classification framework. The methodological exposition follows a progressive structure: we first formalize the core mathematical framework of the SAFD with the BP algorithm. Section III-B detail the Kalman-filter-driven hyperspectral multiscale pyramid architecture. Simultaneously, we enhance the robustness of local textures through LBP processing, ultimately deriving rotation-invariant LBP features. Fig. 1 shows a schematic of the proposed SAFD network with BP structure for feature extraction of HSI. Finally, we construct a semi-supervised graph learning classification algorithm.

A. Proposed SAFD Network With BP

An HSI of size $w \times h \times q$ is 3-D cube data with large spatial dimensions and numerous spectral bands. To improve the processing efficiency of SAFD for HSI, we partition the HSI into nonoverlapping image patches. If the size of each image patch is $w_0 \times h_0 \times q$, then there are (wh/w_0h_0) image patches. Although SAFD is capable of processing 1-D signals, these image patches are multidimensional data. For multidimensional image patches, they can be processed channel by channel. For one image patch, a certain channel c $(c = 1, 2, \ldots, q)$ of the image patch is converted into a 1-D RS, denoted as g_c^i $(i = 1, 2, \ldots, (wh/w_0h_0))$, then each RS with the decomposition level C can be decomposed as

$$g_c^i(z) = \sum_{j=1}^{C} \frac{1}{\alpha} \left\lfloor \alpha \left\langle g_{c,j}^i, e_{a_{c,j}} \right\rangle \right\rfloor \mathcal{B}_{c,j}(z) \tag{8}$$

where

$$\mathcal{B}_{c,j}(z) = e_{a_{c,j}}(z) \prod_{m=1}^{k-1} \frac{z - a_{c,m}}{1 - \bar{a}_{c,m} z}. \tag{9}$$

Here, "$\lfloor\ \rfloor$" denotes the floor function (rounding down to the nearest integer), while α serves as the quantization parameter for the coefficients. The coefficients $a_{c,j}$ can be determined via the SMSP

$$a_{c,j} = \frac{1}{\rho} \left\lfloor \rho \arg\max \left\{ E_i \left| \left\langle g_{c,j}^i, e_a \right\rangle \right|^2, a \in \mathbb{D} \right\} \right\rfloor \tag{10}$$

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Algorithm 1 FFT_SAFD Algorithm

Input: An HSI $\mathcal{Y}$ with $w \times h \times q$; The size of image patches $w_0 \times h_0 \times q$; Image patch $\boldsymbol{S} \in \mathbb{R}^{\frac{wh}{w_0 h_0} \times w_0 h_0 \times q}$; The decomposition level C.

Output: $\{a_{c,j}\}, t$.

for c=1:q **do**
  for i=1:N **do**
    Convert image patches to RSs $\{g_c^i(z)\}_{i=1}^N$;
    $\hat{g}_{c,1}^i(z) = g_c^i(z)$;
  end for
  Calculate $a_{c,1}$ using equation (10).
end for
for c=1:q **do**
  for j=2:C **do**
    for i=1:N **do**
      $\beta_{c,j-1}(z) = \frac{1-\bar{a}_{c,j-1}z}{z-a_{c,j-1}}$;
      $v_{c,j-1}^i(z) = \frac{1}{\alpha}\lfloor\alpha\langle\hat{g}_{c,j-1}^i, e_{a_{c,j-1}}\rangle\rfloor e_{a_{c,j-1}}(z)$;
      $\hat{g}_{c,j}^i(z) = (\hat{g}_{c,j-1}^i(z) - v_{c,j-1}^i(z))\beta_{c,j-1}(z)$;
    end for
  Calculate $a_{c,j}$ using equation (10).
  end for
end for
Return: $\{a_{c,j}\}, t$.

where ρ acts as the quantization parameter for $a_{c,j}$, while $g_{c,j}^i(z)$ is computed recursively via

$$g_{c,j}^i(z) = \left(g_{c,j-1}^i(z) - \frac{1}{\alpha}\left\lfloor\alpha\left\langle g_{c,j-1}^i, e_{a_{c,j-1}}\right\rangle\right\rfloor \times e_{a_{c,j-1}}(z)\right)\beta_{c,j-1}(z) \quad \forall c \in \{1,2,\ldots,q\} \tag{11}$$

and

$$\beta_{c,j-1}(z) = \frac{1-\bar{a}_{c,j-1}z}{z-a_{c,j-1}} \quad \forall c \in \{1,2,\ldots,q\}. \tag{12}$$

Algorithm 1 summarizes the decomposition process of SAFD, abbreviated as FFT_SAFD algorithm.

Building on the patch-based SAFD described above, we now construct a deep architecture to further refine the signal representations to extract deep features. The quality of subsequent feature maps can be enhanced by adaptively optimizing previous feature maps through a self-supervised adaptation mechanism based on the BP algorithm. Fig. 2 illustrates the schematic of jointly updating the convolution kernel $\mathcal{B}^{\ell-1}$ of the $(\ell-1)$th layer by utilizing feature maps from both the $(\ell-1)$th layer and the ℓth layer. Specifically, given previous layer feature map $\boldsymbol{F}^{\ell-1} \in \mathbb{R}^{w\times h}$ at the $(\ell-1)$th layer generated from the SAFD decomposition, the next layer's output $\boldsymbol{F}^{\ell} \in \mathbb{R}^{w\times h}$ at the ℓth layer is computed through a convolutional transformation with learnable filter $\mathcal{B}^{\ell-1} \in \mathbb{R}^{w_0\times h_0\times C}$ at the $(\ell-1)$th layer, formulated as

$$\boldsymbol{F}^{\ell} = \frac{1}{C}\sum_{j=1}^{C} \boldsymbol{F}^{\ell-1} \otimes \mathcal{B}^{\ell-1}(:,:,j) \tag{13}$$

where the calculation of $\mathcal{B}^{\ell-1}$ is detailed in (14). As shown in (13), the learned filter $\mathcal{B}^{\ell-1}$ critically determines the validity of (13) and consequently governs the quality of subsequent

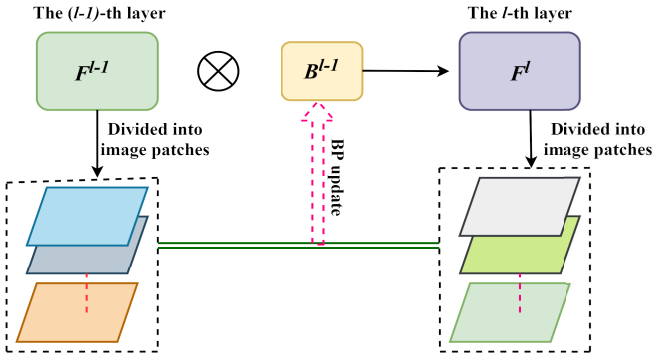


Fig. 2. SAFD deep feature extraction module with BP, where convolution kernel $\mathcal{B}^{\ell-1}$ is updated by utilizing feature maps from both the $(\ell-1)$th layer and the ℓth layer.

hierarchical features. Therefore, $\mathcal{B}^{\ell-1}$ can be optimized via BP. However, the computation of $\mathcal{B}_{c,j}^{\ell-1}$ inherently depends on $\mathbf{a}^{\ell-1}$ in the following equation:

$$\mathcal{B}_{c,j}(z) = e_{a_{c,j}}(z)\prod_{m=1}^{k-1}\frac{z-a_{c,m}}{1-\bar{a}_{c,m}z}. \tag{14}$$

To address this interdependency, we derive the update rules for $\mathbf{a}^{\ell-1}$ and $\mathcal{B}_{c,j}^{\ell-1}$ by calculating the gradient of $\mathbf{a}^{\ell-1}$.

1) BP Update Procedure: Consider layer ℓ with parameter vector $\mathbf{a}^{\ell-1} = [a_1^{\ell-1}, a_2^{\ell-1}, a_3^{\ell-1}, \ldots, a_n^{\ell-1}]^T$. Let $\nabla_{\mathbf{a}^{\ell-1}}\boldsymbol{E} \in \mathbb{C}^{n\times 1}$ denote the gradient with respect to $\mathbf{a}^{\ell-1}$, the learning rate as η, and $v \in \mathbb{C}$ represent the chain rule factor propagated from downstream layers. Through the BP algorithm, we compute componentwise updates for $\mathbf{a}^{\ell-1}$ and aggregate them to obtain the updated parameter vector $\mathbf{a}^{\ell-1}$, which subsequently triggers the update of parameter $\mathcal{B}^{\ell-1}$. For the kth component of vector $\mathbf{a}^{\ell-1}$, the step-by-step derivation is as follows:

$$b_k^{\ell-1} = a_k^{\ell-1} - \eta\cdot\nabla_{a_k^{\ell-1}}\boldsymbol{E}_k\cdot\frac{\partial\phi_k}{\partial a_k^{\ell-1}}\cdot v^{\ell-1} \tag{15}$$

where

$$\psi = \frac{\left|b_k^{\ell-1}\right|^2 b_k^{\ell} - b_k^{\ell} + b_k^{\ell-1} - b_k^{\ell-1}\left(b_k^{\ell}\right)^2}{1-\left|b_k^{\ell-1}b_k^{\ell}\right|^2}$$

$$\frac{\partial\phi_k}{\partial a_k^{\ell-1}} = \mathrm{conj}\left(\frac{1-|\psi|^2}{\left(1-\mathrm{conj}\left(\psi\right)b_k^{\ell-1}\right)^2}\right)$$

$$v^{\ell} = v^{\ell-1}\cdot\prod_{k=1}^{n}\frac{\partial\phi_k}{\partial a_k^{\ell-1}} \tag{16}$$

where the notation conj$(\cdot)$ represents the conjugate. The parameter b_k^{ℓ} in layer ℓ can be updated using the following equation:

$$b_k^{\ell} = b_k^{\ell-1} - \eta\cdot\nabla_{a_k^{\ell-1}}\boldsymbol{E}_k, \quad a_k^{\ell} = b_k^{\ell-1}. \tag{17}$$

All components $b_k^{\ell-1}$ and $a_k^{\ell-1}$ can be iteratively computed by reusing (15) and (17). For the kth $(k \geq 2)$ row of $\mathcal{B}^{\ell-1}$, it is derived by the following formula:

$$\mathcal{B}_k^{\ell-1} = \frac{\sqrt{1-|a_k^{\ell-1}|^2}}{1-\mathrm{conj}\left(a_k^{\ell-1}\right)e^{it}}\cdot\frac{e^{it}-a_{k-1}^{\ell-1}}{\sqrt{1-\left|a_{k-1}^{\ell-1}\right|^2}}\cdot\mathcal{B}_{k-1}^{\ell-1}. \tag{18}$$

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For the first row of $\mathcal{B}^{\ell-1}$, it is derived by the following formula:

$$\mathcal{B}_1^{\ell-1} = \frac{\sqrt{1-|a_1^{\ell-1}|^2}}{1-\operatorname{conj}\left(a_1^{\ell-1}\right)e^{it}}. \tag{19}$$

Through this approach, $\mathcal{B}^{\ell-1}$ can be updated layer by layer to extract deep features.

However, filter $\mathcal{B}^{\ell-1}$, feature $\boldsymbol{F}^{\ell-1}$, and feature $\boldsymbol{F}^{\ell}$ are not isolated; instead, they are interconnected through a self-supervised mechanism that jointly adjusts the features of adjacent layers. Specifically, the features from both previous and next layers are incorporated into the gradient $\nabla_{\mathbf{a}^{\ell-1}}\boldsymbol{E}$ to adapt the parameter vector $\mathbf{a}^{\ell-1}$, which in turn refines $\mathcal{B}^{\ell-1}$ accordingly. The kth component of gradient $\nabla_{\mathbf{a}^{\ell-1}}\boldsymbol{E}$ is calculated as follows:

$$\begin{aligned}
&\nabla_{a_k^{\ell-1}}\boldsymbol{E} \\
&= \boldsymbol{f}_k^{\ell}\left(a_k^{\ell-1}\operatorname{conj}\left(\boldsymbol{f}_k^{\ell}\right)+\left(\left|a_k^{\ell-1}\right|^2-1\right)\operatorname{conj}\left(\frac{\partial \boldsymbol{f}_k^{\ell}}{\partial a_k^{\ell-1}}\right)\right) \\
&\quad -\operatorname{conj}\left(a_k^{\ell-1}\right)\operatorname{conj}\left(\boldsymbol{f}_k^{\ell}\right)\boldsymbol{f}_k^{\ell-1}+\left(1-\left|a_k^{\ell-1}\right|^2\right) \\
&\quad \times\left(\operatorname{conj}\left(\frac{\partial f_k^{\ell}}{\partial a_k^{\ell-1}}\right)\boldsymbol{f}_k^{\ell-1}+\operatorname{conj}\left(\boldsymbol{f}_k^{\ell}\right)\frac{\partial \boldsymbol{f}_k^{\ell-1}}{\partial a_k^{\ell-1}}\right) \\
&\quad +\operatorname{conj}\left(-\operatorname{conj}\left(a_k^{\ell-1}\right)\operatorname{conj}\left(\boldsymbol{f}_k^{\ell}\right)\boldsymbol{f}_k^{\ell-1}+\left(1-\left|a_k^{\ell-1}\right|^2\right)\right. \\
&\quad \left.\times\left(\operatorname{conj}\left(\frac{\partial f_k^{\ell}}{\partial a_k^{\ell-1}}\right)\boldsymbol{f}_k^{\ell-1}+\operatorname{conj}\left(\boldsymbol{f}_k^{\ell}\right)\frac{\partial \boldsymbol{f}_k^{\ell-1}}{\partial a_k^{\ell-1}}\right)\right)
\end{aligned} \tag{20}$$

where

$$\begin{aligned}
\frac{\partial \boldsymbol{f}_k^{\ell}}{\partial a_k^{\ell-1}} &= \frac{\left\langle \hat{\boldsymbol{f}}_k^{\ell}, \frac{\partial e_{a_k^{\ell-1}}(z)}{\partial a_k^{\ell-1}}\right\rangle}{\sqrt{1-\left|a_k^{\ell-1}\right|^2}} \\
\frac{\partial e_{a_k^{\ell-1}}(z)}{\partial a_k^{\ell-1}} &= \frac{z}{(1-\operatorname{conj}(a_k^{\ell-1})z)^2} \\
\hat{\boldsymbol{f}}_k^{\ell} &= (\boldsymbol{f}_{k-1}^{\ell}-\tilde{\boldsymbol{f}}_{k-1}^{\ell}e_a(a_{k-1}^{\ell},e^{it}))\varphi_a(a_{k-1}^{\ell},e^{it}) \\
\tilde{\boldsymbol{f}}_{k-1}^{\ell} &= \frac{\left\langle \hat{\boldsymbol{f}}_k^{\ell}, e_a(a_k^{\ell},e^{it})\right\rangle}{\sqrt{1-\left|a_k^{\ell}\right|^2}}, \quad \hat{\boldsymbol{f}}_1^{\ell}=\boldsymbol{f}^{\ell} \\
e_a(a_{k-1}^{\ell},e^{it}) &= \frac{\sqrt{1-\left|a_{k-1}^{\ell}\right|^2}}{1-\operatorname{conj}(a_{k-1}^{\ell})e^{it}} \\
\varphi_a(a_{k-1}^{\ell},e^{it}) &= \frac{e^{it}-a_{k-1}^{\ell}}{\sqrt{1-\left|a_{k-1}^{\ell}\right|^2}}.
\end{aligned} \tag{21}$$

Here, $\boldsymbol{f}^{\ell}$ denotes the vectorized data obtained by partitioning the feature map $\boldsymbol{F}^{\ell}$ at the ℓth layer into nonoverlapping regions. Algorithm 2 describes the feature extraction procedure for the SAFD network incorporating the BP algorithm.

B. Multiscale Pyramid Construction Based on Kalman Filter for HSIs

Kalman filtering, originally suitable for processing 1-D signals, needs to be extended to multidimensional models to handle HSI data and construct multiscale HSI pyramids. The implementation involves three key phases: 1-D $\longrightarrow$ 2-D extension (developing a 2-D processing framework); 2-D $\longrightarrow$ 3-D extension (establishing a 3-D spatial–spectral model); and ultimately 3-D pyramid construction (creating a hierarchical multiresolution representation). The detailed workflow is as follows.

Algorithm 2 SAFD Network With BP Algorithm

Input:
Image tensor $\mathcal{X} \in \mathbb{R}^{w\times h\times 1\times q}$;
Hyperparameters: Decomposition layers L, SAFD decomposes count m, patch size $w_0\times h_0$, learning rate η and chain rule factor $\nu = 1$.
Output: Deep feature `feat` $\longleftarrow \{\boldsymbol{F}^{\ell}\}$.
for layer $l \leftarrow 1$ to L **do**
  $b_k \leftarrow$ zeros(l, m);
  `feat` $\leftarrow$ zeros$(l+1, w, h)$;
  for image $i \leftarrow 1$ to n **do**
    Current image: `feat`$(1,:,:) \leftarrow \mathcal{X}(:,:,1,i)$;
    for layer $li \leftarrow 1$ to l **do**
      Extract patches $\boldsymbol{S}$ and compute $b_k(li,:)$ by $[b_k(li,:),\sim,t] \leftarrow$ FFT_SAFD$(\boldsymbol{S}, \mathrm{m}-1, \mathrm{m})$ in Algorithm 1;
      Compute $\mathcal{B}^{li}$ using equations (18) and (19);
      Compute `feat`$(li+1,:,:) \leftarrow \boldsymbol{F}^{li}$ by (13);
    for layer $li \leftarrow l$ down to 1 **do**
      Extract image patches from `feat`(li,:,:) and `feat`(li+1,:,:), and vectorize them to get $\boldsymbol{f}^{li}$ and $\boldsymbol{f}^{li+1}$;
      Compute gradient ∇E_j in (20);
      Update parameters: $b_k(li,:)$ in (17);
      Compute derivatives $\frac{\partial \phi_k}{\partial a_k^{\ell-1}}$ in (16);
      Update parameters: $b_k(li-1,:)$ in (15);
      Update chain rule factor ν in (16);
      Compute $\mathcal{B}^{li-1}$ using equations (18) and (19).
return `feat` $\longleftarrow \{\boldsymbol{F}^{\ell}\}$, $\mathcal{B}^{\ell}$.

1) 1-D Kalman Filter Formulation: For a 1-D signal $z = [z_1, z_2, z_3, \ldots, z_w]^{\mathrm{T}} \in \mathbb{R}^{w\times 1}$, the mathematical representation of the Kalman filter prediction stage (for $k \geq 2$) is

$$\begin{aligned}
\hat{x}_k^- &= \hat{x}_{k-1}, \quad \hat{x}_1^- = z_1 \\
P_k^- &= P_{k-1}+Q, \quad P_1 = 1.
\end{aligned} \tag{22}$$

The expression of the update phase (for $k \geq 2$) is

$$\begin{aligned}
K_k &= \frac{P_k^-}{P_k^-+R} \\
\hat{x}_k &= \hat{x}_k^- + K_k\left(z_k-\hat{x}_k^-\right) \\
P_k &= (1-K_k)P_k^-
\end{aligned} \tag{23}$$

where $\hat{x}_k$ is the estimated state, P_k is the error covariance, Q is the process noise, P_k^- represents the prior estimation error covariance, and R is the measurement noise. Then, the result of z after Kalman filtering is $\hat{z}_{\text{filter}} = [z_1, \hat{x}_2, \hat{x}_3, \ldots, \hat{x}_w]^{\mathrm{T}} \in \mathbb{R}^{w\times 1}$. Since the 1-D Kalman filter serves as the foundation for 2-D and 3-D Kalman filters, we will denote the 1-D Kalman filter as $\mathcal{K}_{1\text{-D}}$ to facilitate the derivation of its higher dimensional counterparts. After introducing the notation $\mathcal{K}_{1\text{-D}}$, given a

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1-D signal z, the 1-D Kalman filtering result is $\mathcal{K}_{\text{1-D}}(z) = [z_1, \hat{x}_2, \hat{x}_3, \ldots, \hat{x}_w]^{\text{T}}$.

2) 1-D ⟶ 2-D Extension for 2-D Kalman Filter Case: For 2-D signals, processing can be achieved by sequentially applying filtering operations to each row and column, where every row or column is treated as a 1-D signal. This approach enables effective handling of 2-D signals. Specifically, given a 2-D matrix

$$\mathbf{Z} = \begin{bmatrix} z_{11} & \cdots & z_{1h} \\ \vdots & \ddots & \vdots \\ z_{w1} & \cdots & z_{wh} \end{bmatrix} \in \mathbb{R}^{w \times h}.$$

We fix the columns of $\mathbf{Z}$ and apply 1-D Kalman filtering to each row of $\mathbf{Z}$. That is, for any row i, the result of the ith row signal $z_i = [z_{i1}, z_{i2}, z_{i3}, \ldots, z_{ih}]^{\text{T}} (i = 1, 2, 3 \ldots, w)$ after 1-D Kalman filtering is $\mathcal{K}_{\text{1-D}}(z_i)$. After all rows have been processed, a new row-processed Kalman filter matrix

$$\mathcal{K}_{\text{row}} = \begin{bmatrix} \mathcal{K}_{\text{1-D}}(z_1)^{\text{T}} \\ \vdots \\ \mathcal{K}_{\text{1-D}}(z_w)^{\text{T}} \end{bmatrix} = \begin{bmatrix} z_{11} & \hat{z}_{12} & \cdots & \hat{z}_{1h} \\ \vdots & \vdots & \ddots & \vdots \\ z_{w1} & \hat{z}_{w2} & \cdots & \hat{z}_{wh} \end{bmatrix} \in \mathbb{R}^{w \times h}$$

is obtained. Since the first column is used as the initial prediction point when processing by row, this column does not obtain a 1-D Kalman filter. Therefore, $\mathcal{K}_{\text{row}}$ needs to be subjected to a 1-D Kalman filter again according to its column. That is, for any column j, the result of the jth column signal $z_j = [\hat{z}_{1j}, \hat{z}_{2j}, \hat{z}_{3j}, \ldots, \hat{z}_{wj}]^{\text{T}}$ $(j = 2, 3, 4 \ldots, h)$ after 1-D Kalman filtering is $\mathcal{K}_{\text{1-D}}(z_j)$. After all columns have been processed, a new column-processed Kalman filter matrix

$$\mathcal{K}_{\text{col}} = \left[\mathcal{K}_{\text{1-D}}(z_1), \ldots, \mathcal{K}_{\text{1-D}}\left(z_j\right), \ldots, \mathcal{K}_{\text{1-D}}(z_h)\right] \in \mathbb{R}^{w \times h}.$$

In particular, for $j = 1$, we have $z_1 = [z_{11}, z_{21}, z_{31}, \ldots, z_{w1}]^{\text{T}}$. To simplify the above process, we introduce the notation $\mathcal{K}_{\text{2-D}}$, given a 2-D matrix $\mathbf{Z}$, the 2-D Kalman filtering result is $\mathcal{K}_{\text{2-D}}(\mathbf{Z}) = \mathcal{K}_{\text{col}}(\mathcal{K}_{\text{row}}(\mathbf{Z}))$.

3) 2-D ⟶ 3-D Extension for 3-D Kalman Filter Case: Given a 3-D hyperspectral cube $\mathbf{Y} \in \mathbb{R}^{w \times h \times q}$, for any given spectral band $b \in \{1, 2, \ldots, q\}$, the Kalman filter can perform sequential row-wise and column-wise filtering operations on $\mathbf{Y}_{:,:,b}$, and the filtered result $\hat{\mathbf{Y}}_{:,:,b} = \mathcal{K}_{\text{2-D}}(\mathbf{Y}_{:,:,b})(b = 1, 2, 3, \ldots, q)$ for this band can be derived.

4) Multiscale Pyramid Construction: For any given scale α and the feature representation $\mathbf{Y}^{(\alpha-1)}$ at scale $\alpha - 1$, the multiscale pyramid information at scale α can be derived using the following equation:

$$\begin{aligned} &\hat{\mathbf{Y}}^{(\alpha)} = \left[\mathcal{K}_{\text{col}}\left(\mathcal{K}_{\text{row}}\left(\mathbf{Y}^{(\alpha-1)}_{:,:,b}\right)\right)\right]_{b=1}^{q} \\ &\mathbf{R}^{(\alpha)} = \mathbf{Y}^{(\alpha-1)} - \hat{\mathbf{Y}}^{(\alpha)} \\ &\mathbf{Y}^{(\alpha)} = \mathcal{D}_{0.5}\left(\hat{\mathbf{Y}}^{(\alpha)}\right) \\ &\mathcal{D}_{0.5} : \mathbb{R}^{I_1 \times I_2 \times I_3} \longrightarrow \mathbb{R}^{\lfloor I_1/2 \rfloor \times \lfloor I_2/2 \rfloor \times \lfloor I_3/2 \rfloor} \\ &\mathbf{P}^{(\alpha)} = \mathcal{U}_{\uparrow 2^{\alpha-1}}\left(\mathbf{R}^{(\alpha)}\right) \\ &\mathcal{U}_{\uparrow 2^{\alpha-1}} : \mathbb{R}^{\lfloor J_1/2 \rfloor \times \lfloor J_2/2 \rfloor \times \lfloor J_3/2 \rfloor} \longrightarrow \mathbb{R}^{J_1 \times J_2 \times J_3}. \end{aligned} \tag{24}$$

Final multiscale representation for pyramid aggregation by

$$\mathbf{P}_{\text{multi}} = \left[\text{reshape}\left(\mathbf{P}^{(\alpha)}, \left[wh,\ q\right]\right)\right] \in \mathbb{R}^{wh \times \alpha q}. \tag{25}$$

C. Spatial–Spectral Feature Extraction Based on LBP for HSIs

In HSI classification, the spatial–spectral features are conducive to improving classification performance. To enhance robustness against rotational variations in textures, we incorporate LBP processing, deriving rotation-invariant LBP features. Specifically, we use LBP to extract spatial features from HSIs. Since the spectral information of HSIs is highly redundant, we use PCA for dimensionality reduction. Specific practices are as follows. Given an original hyperspectral cube $\mathbf{Y} \in \mathbb{R}^{w \times h \times q}$, $\mathbf{X} = \text{reshape}(\mathbf{Y}, [wh \times q]) \in \mathbb{R}^{wh \times q}$, then, the covariance of $\mathbf{X}$ is calculated and singular value decomposition (SVD) of this covariance matrix $\mathbf{\Sigma}$ is performed by the following formula:

$$\mathbf{\Sigma} = \frac{1}{wh}\mathbf{X}^{\top}\mathbf{X}, \quad \mathbf{\Sigma} = \mathbf{\Psi}\mathbf{Z}\mathbf{V}^{\top}. \tag{26}$$

The first t principal components of $\mathbf{\Psi}$ are retained and used them for dimensionality reduction

$$\mathbf{X}_{\text{PCA}} = \mathbf{X}\mathbf{\Psi}_t \in \mathbb{R}^{wh \times t}, \quad \mathbf{\Psi}_t = \mathbf{\Psi}(:, 1:t). \tag{27}$$

Next, we use LBP to extract spatial features from $\mathbf{X}_{\text{PCA}}$, which can be defined as

$$\text{LBP}(i, j) = \begin{cases} \displaystyle\sum_{p=0}^{d^2-1} s_p \cdot 2^p & \text{if } U \le 2 \\ d^2 + 1 & \text{otherwise} \end{cases} \tag{28}$$

where s_p is the indicator function, which equals 1 if the condition within the braces is satisfied and 0 otherwise. d indicates the local area window size. $U = |s_{d^2-1} - s_0| + \sum_{p=1}^{d^2-1} |s_p - s_{p-1}|$ represents the number of transitions in the binary pattern. The results based on the block-wise histogram are as follows:

$$\mathbf{H}(i, j, b) = \sum_{(u,v) \in \mathcal{W}} \delta_{\text{LBP}(i+u, j+v), b} \quad (b = 0, 1, \ldots, D-1) \tag{29}$$

where the number of histogram bins $D = d^2(d^2-1)+3$ is determined by the uniform pattern, $\mathcal{W} = \{-d, \ldots, d\} \times \{-d, \ldots, d\}$ is a window neighborhood of size $(2d+1)^2$, $\delta_{x,y}$ is the Kronecker delta function, which equals 1 if $x = y$ and 0 otherwise. $\mathbf{H}$ is 3-D, and we need to reshape it into a 2-D matrix $\hat{\mathbf{H}} = \text{reshape}(\mathbf{H}, [wh, D])$.

Therefore, spatial features can be obtained by

$$\mathbf{F}_{\text{spat}} = \frac{\hat{\mathbf{H}} - \min\left(\hat{\mathbf{H}}\right)}{\max\left(\hat{\mathbf{H}}\right) - \min\left(\hat{\mathbf{H}}\right)} \tag{30}$$

and spectral features can be obtained by

$$\mathbf{F}_{\text{spec}} = \frac{\mathbf{X} - \min(\mathbf{X})}{\max(\mathbf{X}) - \min(\mathbf{X})}. \tag{31}$$

Fusion of $\mathbf{F}_{\text{spat}}$ and $\mathbf{F}_{\text{spec}}$ can obtain spatial–spectral features $\mathbf{F}_{\text{fusion}} = [\mathbf{F}_{\text{spat}}, \mathbf{F}_{\text{spec}}] \in \mathbb{R}^{wh \times (D+q)}$.

D. Complexity Analysis of Feature Extraction

During the feature extraction stage for HSI, the process is divided into three phases, namely, deep feature extraction, Kalman pyramid, and spatial–spectral feature extraction based on LBP. By analyzing the computational complexity of each phase, we can determine the overall computational complexity of the entire feature extraction process.

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1) Phase 1 Complexity Analysis: The first phase involves deep feature extraction, specifically utilizing SAFD with BP to extract deep features. The most time-consuming part of this stage is first computing the convolution kernel $\mathcal{B}^{\ell-1}$ at the $(\ell-1)$th layer using SAFD accelerated by the fast Fourier transform (FFT), followed by performing convolution operations with it. Subsequently, the convolution kernel $\mathcal{B}^{\ell-1}$ is adjusted in a self-supervised manner (via the BP algorithm) using feature maps from both $\boldsymbol{F}^{\ell-1}$ and $\boldsymbol{F}^{\ell}$. For input tensor $\mathcal{X} \in \mathbb{R}^{w\times h\times q}$ with decomposition level C, number of layer ℓ and patch size $w_0 \times h_0$. The overall complexity of the first phase is dominated by three components: the FFT_SAFD algorithm contributing $\mathcal{O}((m-1)[mw_0h_0\log(w_0h_0)+wh\log(w_0h_0)])$, arising from FFTs on overlapping $w_0\times h_0$ patches across the $w\times h$ spatial domain; the convolution operation with $\mathcal{O}(\ell mwhw_0h_0)$, reflecting ℓ layers of spatial convolution with $w_0\times h_0$ kernels; and the BP parameter update costing $\mathcal{O}(m^2w_0h_0)$, representing gradient computation and weight updates per patch element. Complexity scales linearly with spatial dimensions (w,h), layers (ℓ), and decomposition level (m), but exhibits super-linear behavior via FFT logarithms and quadratic m-dependence in BP. To sum up, the computational complexity of the first phase is

$$\mathcal{O}_{\text{phase_first}} = \mathcal{O}\left(\underbrace{(m-1)\left[mw_0h_0\log(w_0h_0)+wh\log(w_0h_0)\right]}_{\text{FFT_SAFDalgorithm}} + \underbrace{\ell mwhw_0h_0}_{\text{Convolution operation}} + \underbrace{m^2w_0h_0}_{\text{BP parameter update}}\right). \tag{32}$$

2) Phase 2 Complexity Analysis: For input tensor $\mathcal{X} \in \mathbb{R}^{w\times h\times q}$, and the pyramid decomposition level is α. For an α-layer pyramid where each level is down-sampled to 1/4 of the spatial resolution of its predecessor (halved in both width and height), the total computational complexity aggregates layer-wise complexities into a geometric series. This arises because the number of pixels (or operations) per layer reduces by a factor of 1/4 at each scale, yielding a complexity summation

$$\begin{aligned}\mathcal{O}_{\text{phase_second}} &= \sum_{l=1}^{\alpha}\frac{whq}{4^{l-1}}, \quad (\alpha\longrightarrow\infty)\\ &= \mathcal{O}\left(\frac{4}{3}whq\right)\end{aligned} \tag{33}$$

and the computational complexity of the reconstruction phase (upsampling) is also $\mathcal{O}((4/3)whq)$. Therefore, the overall complexity of the multiscale pyramid construction stage is $\mathcal{O}((8/3)whq)$, and $\mathcal{O}(whq)$ is obtained by ignoring the constant (8/3) of $\mathcal{O}((8/3)whq)$.

3) Phase 3 Complexity Analysis: The third phase involves spatial–spectral feature extraction. Since the spectral dimension information of HSI is highly redundant, when extracting spatial–spectral features, the PCA method based on SVD is used for dimensionality reduction. Given the number of histogram bins D and the local area window size d, the most time-consuming part is the covariance for PCA, SVD, and LBP. The third phase's computational complexity is dominated by three core operations: PCA costing $\mathcal{O}(q^2wh)$ for covariance matrix construction across q feature channels over $w\times h$ spatial positions; SVD requiring $\mathcal{O}(q^3)$ to decompose the $q\times q$ covariance matrix; and LBP feature extraction demanding $\mathcal{O}(whq(d^2+(2d+1)^{2\text{-D}}))$ for local pattern computation and histogram encoding within dd-sized neighborhoods using DD bins. Complexity scales quadratically with channel dimension q (PCA), cubically for SVD, and linearly with spatial area wh, while LBP is sensitive to both neighborhood size dd and histogram resolution D. To sum up, the computational complexity of the third phase is

$$\mathcal{O}_{\text{phase_third}} = \mathcal{O}\left(\underbrace{q^2wh}_{\text{PCA}} + \underbrace{q^3}_{\text{SVD}} + \underbrace{whq\left(d^2+(2d+1)^{2\text{-D}}\right)}_{\text{LBP}}\right). \tag{34}$$

In summary, the overall computational complexity of the feature extraction stage of the proposed method is the sum of the computational complexity of the three phases, namely,

$$\mathcal{O}_{\text{total}} = \mathcal{O}_{\text{phase_first}} + \mathcal{O}_{\text{phase_second}} + \mathcal{O}_{\text{phase_third}}. \tag{35}$$

E. Classification for HSI

The feature $\mathbf{F}_{\text{SAFD}} = \text{reshape}(feat,[wh,L+1])$ in Algorithm 2 extracted by SAFD with BP, multiscale pyramid features $\mathbf{P}_{\text{multi}}$, and spatial–spectral feature $\mathbf{F}_{\text{fusion}}$ are fused to form a new hybrid feature matrix $\mathcal{F}_{\text{hyb}} = [\mathbf{F}_{\text{SAFD}}\ \ \mathbf{P}_{\text{multi}}\ \ \mathbf{F}_{\text{fusion}}]$.

In this section, we will classify the extracted combined features $\mathcal{F}_{\text{hyb}} \in \mathbb{R}^{wh\times((\alpha+1)q+L+1+D)}$, and normalize it

$$\hat{\mathcal{F}}^i_{\text{hyb}} = \frac{\mathcal{F}^i_{\text{hyb}} - \text{mean}\left(\mathcal{F}^i_{\text{hyb}}\right)}{\text{var}\left(\mathcal{F}^i_{\text{hyb}}\right)} \quad \forall\ i\in\{1,2,\ldots,(\alpha+1)q+L+1+D\} \tag{36}$$

where $\hat{\mathcal{F}}^i_{\text{hyb}} \in \mathbb{R}^{wh\times 1}$ denotes the normalized data, and mean$(\cdot)$ and var$(\cdot)$ represent the mean and variance, respectively.

GSL in [5] is used to classify $\hat{\mathcal{F}}_{\text{hyb}}$. First, $\hat{\mathcal{F}}_{\text{hyb}}$ is divided into training samples $\mathcal{X}_{\text{tr}} = [\boldsymbol{x}_1,\boldsymbol{x}_2,\ldots,\boldsymbol{x}_r]$ and test samples $\mathcal{X}_{te} = [\boldsymbol{x}_{r+1},\boldsymbol{x}_{r+2},\ldots,\boldsymbol{x}_M]$, where r is the number of training samples. GSL can be solved by the following quadratic optimization problem with k-class:

$$\min_{\mathcal{T}} \text{tr}\left(\mathcal{T}-\mathcal{V}\right)^T\mathcal{M}\left(\mathcal{T}-\mathcal{V}\right)+\text{tr}\left(\mathcal{T}^T\mathcal{G}\mathcal{T}\right) \tag{37}$$

where tr$(\cdot)$ is the trace function, $\mathcal{T}\in\mathbb{R}^{M\times k}$ contains predicted labels for test samples in its corresponding rows

$$\mathcal{V} = \begin{pmatrix} v_{11} & \cdots & v_{1k}\\ \vdots & \ddots & \vdots\\ v_{M1} & \cdots & v_{Mk}\end{pmatrix}, \quad \begin{cases} v_{ij}=1 & \text{if}\ \ \boldsymbol{x}_i\in j\\ v_{ij}=0 & \text{otherwise.}\end{cases} \tag{38}$$

$\mathcal{M}\in\mathbb{R}^{M\times M}$ is defined as follows:

$$\mathcal{M} = \begin{pmatrix} \theta_1 & \cdots & 0\\ \vdots & \ddots & \vdots\\ 0 & \cdots & \theta_M\end{pmatrix}, \quad \begin{cases} \theta_i=\delta_n & \text{if}\ \ 1\le i\le n\\ \theta_i=\delta_{M-n} & \text{otherwise}\end{cases} \tag{39}$$

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TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES IN THE IP

Class number	Class name	Training samples	Test samples
1	Alfalfa	5	41
2	Corn-notill	143	1285
3	Corn-mintill	83	747
4	Corn	24	213
5	Grass-pasture	48	435
6	Grass-trees	73	657
7	Grass-pasture-mowed	3	25
8	Hay-windrowed	48	430
9	Oats	2	18
10	Soybean-notill	97	875
11	Soybean-mintill	246	2209
12	Soybean-clean	59	534
13	Wheat	21	184
14	Woods	127	1138
15	Buildings-Grass-Trees-Drives	39	347
16	Stone-Steel-Towers	9	84
	Total	1027	9222

where parameters δ_n and δ_{M-n} need to be preset with initial values. $\mathcal{G} \in \mathbb{R}^{M\times M}$ is graph Laplacian, which is calculated by

$$\mathcal{G} = \mathcal{L} - \vartheta \tag{40}$$

where $\vartheta = \left\{\kappa_{ij}\right\}(i, j = 1, \ldots, M) \in \mathbb{R}^{M\times M}$, and κ_{ij} is defined as follows:

$$\kappa_{ij} = e^{-\frac{\|x_i - x_j\|^2}{2\xi^2}} (i \neq j), \quad \kappa_{ii} = 0 \tag{41}$$

where ξ is the preset parameter, $\mathcal{L}$ is defined by

$$\mathcal{L} = \begin{pmatrix} \phi_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \phi_M \end{pmatrix}, \quad \phi_i = \sum_{j=1}^{M} \kappa_{ij}. \tag{42}$$

IV. EXPERIMENTAL RESULTS AND ANALYSIS

To verify the validity of the proposed method, experiments were carried out on the Indian Pines (IP)[1], Houston2013 (HU)[2], and PaviaU (PU)[3] datasets, respectively.

A. Datasets and Setting

1) Datasets: Three HSI datasets are utilized in our experiments, including the IP, HU, and PU datasets.

a) IP dataset: This HSI dataset was acquired in 1992 by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over an agricultural region in North-Western Indiana, USA. The dataset initially contains 220 spectral bands, but 20 bands were removed due to noise, resulting in 200 usable bands. The spatial resolution is 145×145 pixels, with 16 distinct land cover categories (e.g., alfalfa, corn, and woods). For experimental validation, 10% of the samples were randomly selected for training, while the remaining 90% were reserved for testing, and the numbers of training and test samples are summarized in Table II.

b) HU dataset: Collected by the ITRES CASI-1500 sensor over the University of Houston campus and its adjacent urban areas in Texas, USA, this dataset provides a cloud-free HSI distributed by the IEEE Geoscience and Remote Sensing Society (GRSS). It comprises 144 spectral bands with a spatial dimension of 349×1905 pixels. The scene is annotated with 15 land cover classes, including highway, road, and trees. Following standard practice, 10% of the labeled data was utilized for model training, and the rest served as the test set. The numbers of training and test samples are summarized in Table III.

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES IN THE HU

Class number	Class name	Training samples	Test samples
1	Healthy Grass 1	125	1126
2	Stressed Grass 2	125	1129
3	Synthetic Grass	70	627
4	Trees	124	1120
5	Soil	124	1118
6	Water	33	292
7	Residential	127	1141
8	Commercial	124	1120
9	Road	125	1127
10	Highway	123	1104
11	Railway	124	1111
12	Parking Lot 1	123	1110
13	Parking Lot 2	47	422
14	Tennis Court	43	385
15	Running Track	66	594
	Total	1503	13526

TABLE IV
NUMBER OF TRAINING AND TESTING SAMPLES IN THE PU

Class number	Class name	Training samples	Test samples
1	Asphalt	663	5968
2	Meadows	1865	16784
3	Gravel	210	1889
4	Trees	306	2758
5	Metal sheets	135	1210
6	Bare soil	503	4526
7	Bitumen	133	1197
8	Bricks	368	3314
9	Shadows	95	852
	Total	4278	38498

c) PU dataset: Captured by the Reflective Optics System Imaging Spectrometer (ROSIS) over the University of Pavia, Italy, this HSI dataset consists of 103 spectral bands and spans a spatial area of 610×340 pixels. It encompasses nine urban land cover types, such as asphalt, gravel, and trees. To ensure consistency with other datasets, we adopted a 10%–90% split ratio for training and testing sets, respectively. The numbers of training and test samples are summarized in Table IV.

2) Evaluation Metrics: To comprehensively evaluate the performance of the proposed model, we employ three widely recognized metrics in HSI classification: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ). The rationale for selecting this specific suite of metrics is as follows.

a) Overall accuracy: OA is the most intuitive metric, calculated as the ratio of correctly classified pixels to the total number of test pixels. It provides a general overview of the model's global performance. However, OA can be heavily influenced by class imbalance, where high accuracy on large classes may mask poor performance on smaller ones. Therefore, while OA offers a primary performance snapshot, it is insufficient alone and must be interpreted alongside other metrics.

[1]http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

[2]http://www.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

[3]http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_University_scene

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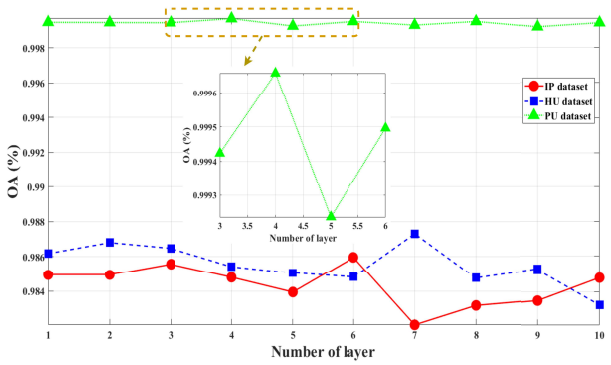


Fig. 3. Impact of the number of network layers on classification accuracy.

b) Average accuracy: AA addresses the limitation of OA in imbalanced datasets. It is computed as the mean of the per-class accuracies (i.e., the producer's accuracies). This metric assigns equal weight to each class, regardless of its sample size. Consequently, AA is a more robust indicator of the model's ability to generalize across all land cover types, ensuring that its performance on rare classes is not overshadowed by dominant classes.

c) Kappa coefficient (κ)*:* The Kappa coefficient is a statistically robust measure that quantifies the agreement between the classification map and the ground truth reference data, while correcting for the agreement expected by random chance. A value of 1 indicates perfect agreement, while 0 indicates agreement no better than random. Kappa is particularly valuable as it accounts for all elements of the confusion matrix, not just the diagonal ones, providing a more reliable and conservative evaluation of classification effectiveness than OA alone.

B. Parameters Analysis and Setting

The main parameters involved in the proposed method include: the number of SAFD network layers with BP algorithm, the SAFD decomposition level, and image patch size; scale parameter α in the multiscale pyramid, and local region window size for LBP. We systematically these hyperparameters through controlled experiments across all datasets. Below is a consolidated analysis summarizing parameter sensitivity and optimal configurations.

1) Number of SAFD Network Layers: The depth of the SAFD network significantly impacts feature extraction and computational efficiency. As shown in Fig. 3, classification accuracy exhibits dataset-dependent trends: for IP and HU: accuracy peaks at six layers (IP, OA: 98.78%) and seven layers (HU, OA: 98.97%), declining beyond these depths. This indicates that intermediate depths optimally balance discriminative power and overfitting; for the PU dataset, accuracy stabilizes above 99% at four layers, with negligible gains beyond this depth, prioritizing computational efficiency. Optimal choice: six layers (IP), seven layers (HU), and four layers (PU) to maximize accuracy while curbing complexity.

2) SAFD Decomposition Level and Image Patch Size: Interactions between decomposition level (m) and patch size (w_0) critically influence spatial–spectral feature quality in Fig. 4. For the IP dataset, the highest accuracy (98.78%) is at $m = 1$ and $w_0 = 5$. Smaller patches (w_0 range from 3 to 9) yield stable results, while larger patches ($w_0 = 17$) occasionally improve accuracy only at high m (e.g., $m = 9$, OA: 98.72%), but inconsistently. For HU dataset, peak accuracy (98.97%) at $m = 8$ and $w_0 = 19$, though medium patches (w_0 range from 7 to 15) generalize better across m. Low m ($m = 1, 2$) also achieves >98.85% accuracy with $w_0 = 7$. For the PU dataset, minimal sensitivity (OA: 99.94%~99.97%). Best performance at $m = 1, 3, 5, 7, 9$ with $w_0 = 3$, though differences are marginal. Therefore, $m = 1$ and $w_0 = 5$ (IP); $m = 8$ and $w_0 = 19$ (HU); and $m = 3$ and $w_0 = 3$ (PU) were selected for peak observed accuracy.

TABLE V
SUMMARY OF OPTIMAL HYPERPARAMETERS PER DATASET

Datasets	Layers	Decomposition (m)	Patch size (w_0)	Scale (α)	LBP window size
IP	6	1	5	8	7
HU	7	8	19	11	3
PU	4	3	3	7	7

3) Scale Parameter α in Multiscale Pyramid: Fig. 5 below illustrates the effect of pyramid scales on multiscale fusion: accuracy peaks at $\alpha = 8$ (98.61%), declining sharply at $\alpha = 12$ (94.77%) on IP dataset, indicating optimal texture capture at mid-scales; best performance at $\alpha = 11$ (98.57%), with larger scales ($\alpha > 10$) generally superior for the HU dataset; near-invariance (OA: 99.92%~99.96%), confirming robustness to scale variations for the PU dataset. Therefore, $\alpha = 8$ (IP), $\alpha = 11$ (HU), and $\alpha = 7$ (PU) were later chosen for balance across datasets.

4) LBP Window Size: Local texture extraction sensitivity is shown in Fig. 6. For the IP dataset, accuracy peaks at window size 7 (98.67%), with medium windows (range from 5 to 9) outperforming extremes. For the HU dataset, maximum accuracy (98.92%) at size 3, declining monotonically with larger windows due to loss of fine textures. For the PU dataset, consistency across sizes (OA: 99.89%~99.94%), peaking at size 7 (99.94%). Therefore, size 7 on IP, size 3 on HU, and size 7 on PU to align with the accuracy maxima.

5) Parameter Configurations: Based on the above analysis, the optimal hyperparameters of each dataset are summarized in Table V.

These configurations maximize classification accuracy (OA) as empirically validated in Figs. 3–6. The tuning process revealed dataset-specific sensitivities, e.g., HU benefits from deeper networks and large patches at high decomposition levels, while PU tolerates wider parameter ranges due to inherent separability. This systematic analysis underscores the necessity of dataset-aware hyperparameter optimization.

C. Classification Results

In this section, we report the classification results of the proposed methods and the comparison results with other state-of-the-art methods. We selected comparative methods that are closely related to the proposed method in terms of classifiers and network architectures, namely, classifier graph-based learning (GSL) [48], three CNN-type methods: CNN proposed by LeCun (LeNet) et al. [49], random patches network

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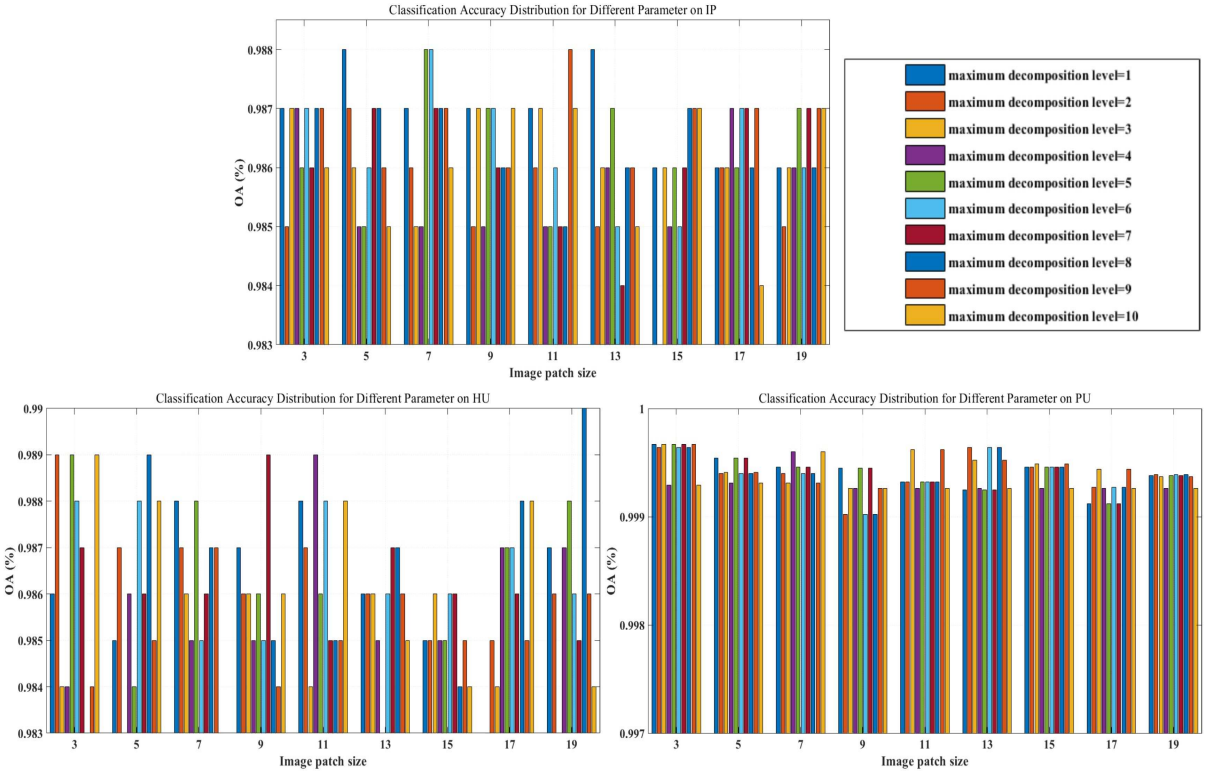


Fig. 4. Effect of image patch size and maximum decomposition level combinations on classification accuracy.

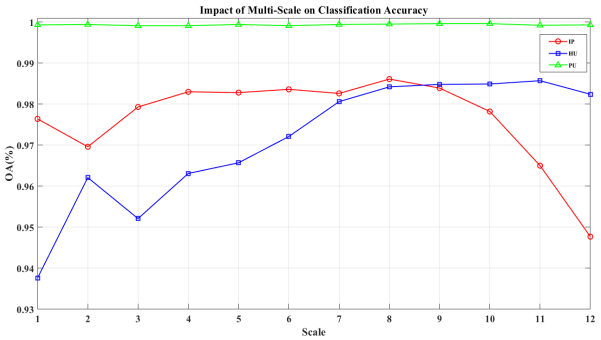


Fig. 5. Impact of multiscale Kalman filter on classification accuracy.

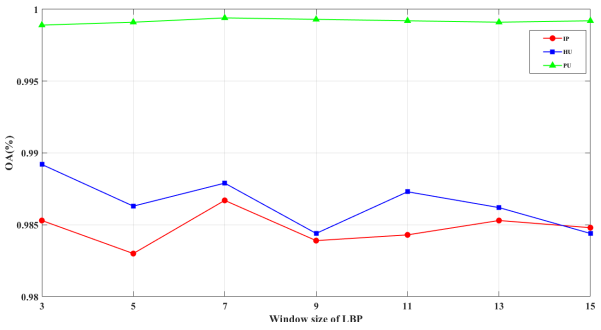


Fig. 6. Effect of LBP window size variations on classification accuracy.

(RPNet) [38] and spatial–spectral RPNet (SSRPNet) [5]; two CNN type methods based on higher order tensor type: DHTC-SCNet [39] and DHTSRNet [40]; and CNN type method based on SAFD (SAFDNet) [46]. Moreover, we choose some state-of-the-art methods to compare with the proposed method, including transformers (such as SS_TMNet [33], DCTN [34], MorphFormer [32], and ACTN [1]). We conduct experiments on three widely used HSI datasets in terms of OA, AA, and κ. The results are reported on Tables VI–VIII. In tables, the best results are marked in bold.

As can be seen from the results in Tables VI–VIII, the proposed method demonstrates superior classification performance across all three datasets (IP, HU, and PU) in terms of OA, AA, and κ. For the IP dataset, ours achieves the highest OA (98.93%), AA (99.03%), and κ (98.78%), outperforming the suboptimal methods SSRPNet (OA: 98.7%) and DHTCSCNet (AA: 98.78%). In the HU dataset, ours maintains dominance with OA (98.88%) and AA (99.06%), significantly surpassing DCTN (OA: 98.31%) and ACTN (AA: 98.16%), while its κ value (98.79%) is comparable to the best-performing ACTN (OA: 98.46%). For the PU dataset, ours achieves near-perfect results (OA: 99.93%, AA: 99.94%, κ: 99.9%), far exceeding other methods (e.g., DHTCSCNet's OA: 99.21%). Notably, traditional methods (e.g., GSL and LeNet) perform poorly across all datasets (e.g., GSL's OA: 72.35% in IP), while recent methods (e.g., Morphformer and ACTN) show instability (e.g., Morphformer's OA drops to 90.98% in HU). These results validate the robustness and comprehensive superiority of the proposed method in cross-dataset classification tasks.

Figs. 7–9 provide a visual comparative analysis of the performance of the proposed method on IP, HU, and PU datasets. From Figs. 7 to 9, it is clear that the proposed method

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TABLE VI
QUANTITATIVE RESULTS COMPARISON AMONG THE DIFFERENT METHODS WITH 10% TRAINING SAMPLES ON IP

Class	GSL	LeNet	RPNet	SSRPNet	DHTCSCNet	DHTSRNet	SAFDNet	SS_TMNet	DCTN	MorphFormer	ACTN	Ours
Alfalfa	96.88	76.34	82.20	98.06	**100**	**100**	**100**	87.48	72.29	82.13	95.45	**100**
Com-notill	59.80	94.85	94.82	98.08	**99.04**	91.76	98.57	88.56	95.70	93.38	94.92	98.40
Corn-mintill	60.51	90.72	93.63	97.39	98.15	93.79	**98.20**	76.50	88.93	91.28	93.09	98.09
Corn	61.49	59.91	89.30	96.08	99.07	96.02	99.17	82.19	92.77	95.26	**99.66**	97.29
Grass-pasture	94.22	94.00	96.92	98.93	99.53	99.26	99.14	81.71	92.04	94.72	94.74	**99.86**
Grass-trees	73.89	99.30	99.22	99.63	97.17	98.98	99.02	97.76	98.64	95.47	96.24	**99.79**
Grass-pasture-mowed	43.59	76.80	86.80	**100**	96.15	**100**	94.72	72.09	73.76	71.56	29.73	97.09
Hay-windrowed	91.49	97.37	99.07	**100**	**100**	99.81	98.40	94.39	97.75	99.81	99.98	**100**
Oats	69.23	41.67	68.89	83.60	**100**	**100**	**100**	68.66	76.70	21.28	0.00	**100**
Soybean-notill	78.00	92.85	93.18	98.61	98.95	97.92	98.60	87.19	93.57	88.80	88.77	97.72
Soybean-mintill	70.61	98.51	97.99	**99.18**	98.79	97.61	97.60	90.70	95.46	96.27	97.01	98.87
Soybean-clean	47.80	87.53	93.09	97.39	95.31	96.84	98.67	81.85	94.68	87.66	93.65	**98.66**
Wheat	76.06	97.77	98.48	98.53	**100**	**100**	99.82	97.18	99.89	94.35	98.03	**100**
Woods	87.88	99.11	99.50	99.65	99.39	99.95	99.93	96.21	98.49	98.87	98.93	**99.95**
Buildings-Grass-Trees-Drives	58.10	91.27	95.68	**100**	98.86	98.08	99.07	63.92	77.76	96.06	98.50	99.71
Stone-Steel-Towers	**100**	19.88	76.07	93.13	**100**	98.76	98.83	87.73	96.80	25.11	67.32	99.06
OA(%)	72.35	93.90	96.09	98.70	98.67	97.05	98.58	84.67	92.85	94.03	95.40	**98.93**
AA(%)	73.10	82.37	91.55	98.24	98.78	98.05	98.73	85.56	86.55	82.75	88.47	**99.03**
κ(%)	68.17	93.00	95.54	98.52	98.48	96.63	98.38	82.66	91.87	93.18	94.75	**98.78**

TABLE VII
QUANTITATIVE RESULTS COMPARISON AMONG THE DIFFERENT METHODS WITH 10% TRAINING SAMPLES ON HU

Class	GSL	LeNet	RPNet	SSRPNet	DHTCSCNet	DHTSRNet	SAFDNet	SS_TMNet	DCTN	Morphformer	ACTN	Ours
Healthy Grass	89.57	98.70	98.77	**99.41**	98.60	94.86	95.97	97.60	98.86	85.97	97.20	99.24
Stressed grass	94.09	98.73	98.85	95.01	95.18	95.44	91.82	98.44	**99.33**	88.97	98.50	99.09
Synthetic GrassTrees	90.91	99.19	**100**	**100**	**100**	**100**	**100**	99.50	99.74	91.37	99.13	**100**
Trees	93.62	97.43	98.79	88.96	94.95	**100**	97.56	97.26	99.21	90.96	96.19	97.65
Soil	84.25	98.97	99.75	98.02	93.90	99.20	98.92	98.19	98.65	97.96	**99.89**	99.18
Water	97.12	94.21	96.88	98.28	**100**	**100**	**100**	93.67	98.75	90.85	93.23	**100**
Residential	64.57	97.77	98.09	88.40	93.71	94.72	**98.66**	94.54	98.20	85.78	97.97	97.10
Commercial	77.80	94.40	95.54	97.71	98.01	99.15	97.43	95.74	98.22	90.76	99.57	**99.98**
Road	60.82	96.50	**97.83**	94.07	96.50	96.41	95.25	94.29	97.66	87.54	97.00	97.65
Highway	71.58	99.09	98.99	99.41	98.40	95.58	86.41	96.91	98.89	93.89	**100**	98.46
Railway	64.44	98.51	99.21	99.46	99.20	98.23	83.17	94.94	98.51	91.69	**99.97**	99.62
Parking Lot 1	60.73	97.04	98.29	98.94	97.63	98.46	94.50	96.50	98.96	88.63	**99.59**	99.43
Parking Lot 2	34.29	71.26	79.69	97.94	99.04	**99.75**	94.19	93.42	97.68	92.25	99.40	98.74
Tennise Court	96.11	99.04	99.56	**100**	99.23	98.47	98.97	99.88	**100**	99.04	**100**	**100**
Running Track	95.48	98.47	99.78	**100**	**100**	**100**	**100**	98.98	99.24	93.32	98.97	99.70
OA(%)	78.06	96.95	97.96	96.41	97.10	97.58	94.48	96.22	98.31	90.98	98.58	**98.88**
AA(%)	78.36	95.95	97.33	97.04	97.62	98.02	95.52	95.15	97.32	90.99	98.16	**99.06**
κ(%)	76.26	96.70	97.79	96.12	96.87	97.39	94.04	95.92	98.17	90.24	98.46	**98.79**

TABLE VIII
QUANTITATIVE RESULTS COMPARISON AMONG THE DIFFERENT METHODS WITH 10% TRAINING SAMPLES ON PU

Class	GSL	LeNet	RPNet	SSRPNet	DHTCSCNet	DHTSRNet	SAFDNet	SS_TMNet	DCTN	Morphformer	ACTN	Ours
Asphalt	84.52	97.94	99.09	98.80	98.30	98.80	98.89	96.11	98.57	94.76	99.37	**99.97**
Meadows	86.55	**99.91**	**99.91**	99.83	99.89	98.62	99.47	92.67	97.66	99.67	99.85	99.88
Gravel	61.75	94.72	98.75	98.72	99.89	95.15	**100**	92.35	90.37	86.82	99.36	**100**
Trees	85.75	98.65	99.12	99.51	99.77	99.70	99.92	96.46	98.69	87.85	96.63	**99.81**
Painted metal sheets	91.65	**100**	99.97	93.39	94.75	99.88	99.92	99.66	99.99	94.58	95.66	99.85
Bare Soil	73.29	99.69	99.84	**100**	**100**	99.12	99.63	99.91	99.04	99.84	99.94	**100**
Bitumen	69.29	96.65	99.00	99.82	97.32	96.88	94.73	99.05	98.86	90.65	99.14	**100**
Self-Blocking Bricks	68.26	97.17	99.10	98.52	98.66	96.92	95.18	98.31	95.04	92.82	99.13	**100**
Shadows	**100**	89.91	99.88	97.97	95.98	99.95	99.88	98.02	99.95	86.53	88.94	99.98
OA(%)	81.93	98.68	99.56	99.24	99.21	98.48	98.94	91.74	96.57	96.33	99.09	**99.93**
AA(%)	80.12	97.18	99.41	98.51	98.28	98.34	98.62	95.31	96.32	96.22	96.81	**99.94**
κ(%)	75.76	98.25	99.42	98.99	98.95	97.98	98.59	89.44	95.49	95.13	98.8	**99.90**

can predict all classes with a high degree of accuracy. More importantly, it generates a classification map that remains faithful to the ground truth, suggesting that the proposed method effectively integrates different features. This could be attributed to the design of the proposed feature extractor, which is specifically engineered to fuse these different types of features coherently, thereby enhancing its classification performance. On the other hand, a careful examination of the figures also reveals that most of the compared methods would produce classification maps with notable levels of noise. This finding is in alignment with the quantitative results summarized in Tables VI–VIII provided above.

D. Analysis and Discussion

Section IV-C have validated that the proposed method achieves satisfactory classification performance under a 10% proportion of training samples. This section will discuss the classification effectiveness of the proposed method across varying small-sample sizes. In addition, ablation studies will be presented to verify the impact of different modules in the proposed method on classification performance. Details are as follows.

1) Impact of Small-Sample Sizes: To validate the classification performance of the proposed method under limited

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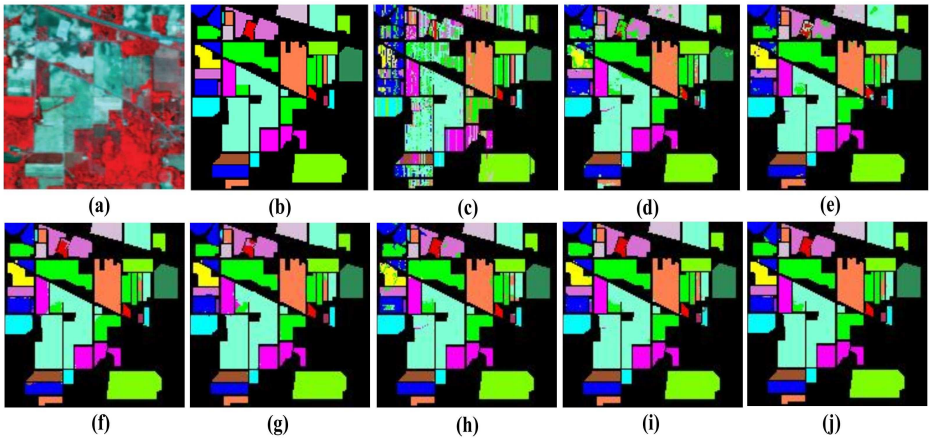


Fig. 7. Classification maps of different methods on IP. (a) False-color map; (b) ground truth; (c) GSL (72.35%); (d) LeNet (93.90%); (e) RPNet (96.09%); (f) SSRPNet (98.70%); (g) DHTCSCNet (98.67%); (h) DHTSRNet (97.05%); (i) SAFDNet (98.58%); and (j) ours (98.93%).

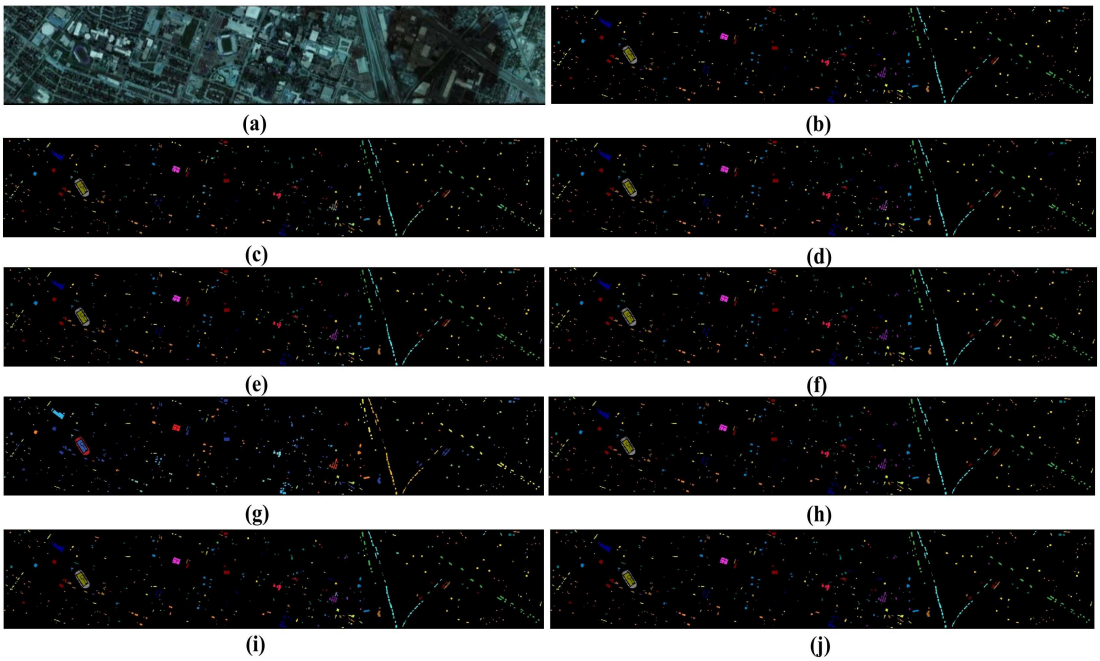


Fig. 8. Classification maps of different methods on HU. (a) False-color map; (b) ground truth; (c) GSL (78.06%); (d) LeNet (96.95%); (e) RPNet (97.96%); (f) SSRPNet (96.41%); (g) DHTCSCNet (97.10%); (h) DHTSRNet (97.58%); (i) SAFDNet (94.48%); and (j) ours (98.88%).

TABLE IX
COMPARISON OF COMPUTATIONAL EFFICIENCY ACROSS DIFFERENT METHODS ON HU DATASET

	GSL	LeNet	RPNet	SSRPNet	DHTCSCNet	DHTSRNet	SAFDNet	SS_TMNet	DCTN	Morphformer	ACTN	Ours
OA(%)	78.06	96.95	97.96	96.41	97.1	97.58	94.48	96.22	98.31	90.98	98.58	**98.88**
AA(%)	78.36	95.95	97.33	97.04	97.62	98.02	95.52	95.15	97.32	90.99	98.16	**99.06**
κ(%)	76.26	96.7	97.79	96.12	96.87	97.39	94.04	95.92	98.17	90.24	98.46	**98.79**
training time (s)	**35.28**	306.54	224.9	320.7	319.43	1512.6	2278.2	333.93	253.16	874.99	870.97	346.41
test time(s)	**0.02**	5.41	4.21	77.92	45.47	8.13	583.5	31.12	20.69	4.16	3.65	2.03
Param (MB)	–	–	–	0.21	0.15	1.79	0.13	83.33	45.32	87.04	14.04	**0.02**

training samples, we conducted experiments on the IP, HU, and PU datasets, investigating the classification accuracy of different methods with 5, 10, 15, 20, 25, and 30 training samples. The experimental results are illustrated in Fig. 10. It is important to note that the comparative methods selected in the figure are those most similar to the proposed method in terms of network architecture and classifier design. The experimental results demonstrate that the classification accuracy (OA%) of all methods on the IP, HU, and PU datasets significantly improves as the number of training samples increases from 5 to 30. Notably, the proposed method exhibits a clear advantage in small-sample scenarios. For instance, with only five training

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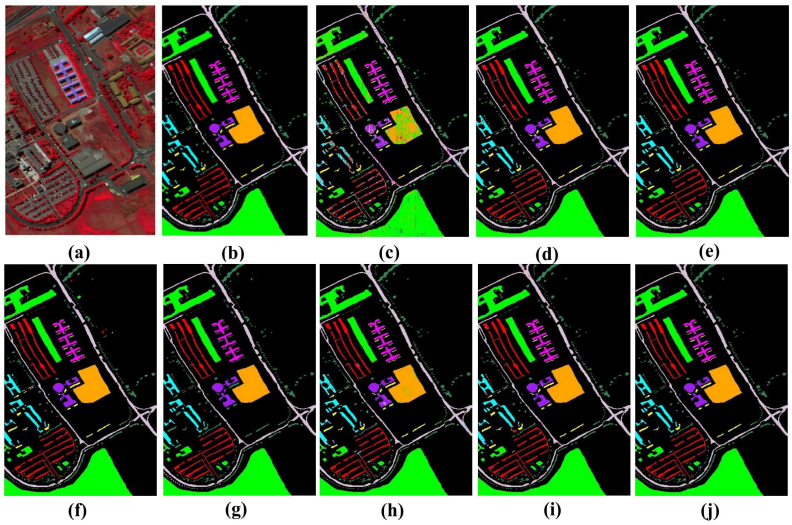


Fig. 9. Classification maps of different methods on PU. (a) False-color map; (b) ground truth; (c) GSL (81.93%); (d) LeNet (98.68%); (e) RPNet (99.56%); (f) SSRPNet (99.24%); (g) DHTCSCNet (99.21%); (h) DHTSRNet (98.48%); (i) SAFDNet (98.94%); and (j) ours (99.93%).

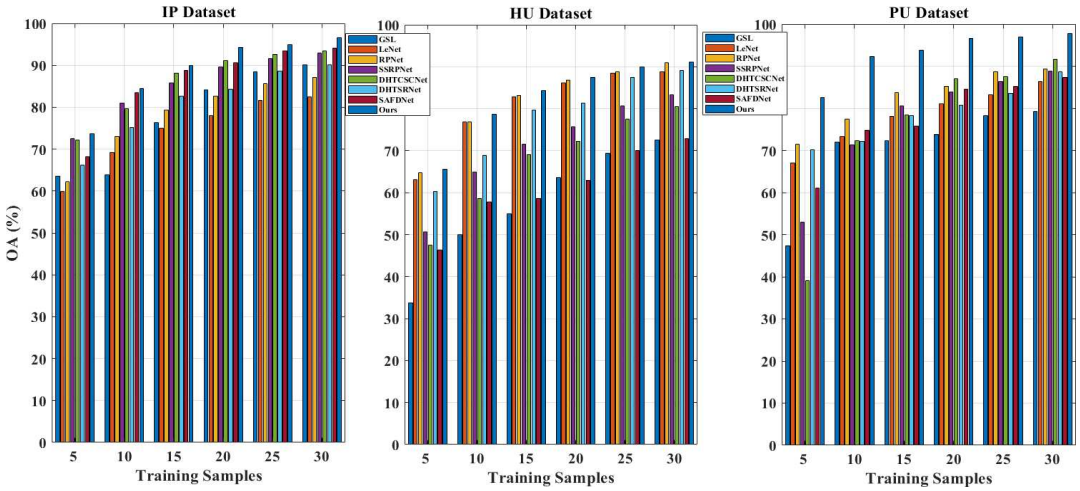


Fig. 10. Impact of small-sample sizes on IP, HU, and PU datasets.

TABLE X
IMPACT OF EACH PART AND COMBINATIONS OF DIFFERENT PARTS ON CLASSIFICATION PERFORMANCE, WHERE THE SYMBOL "+" INDICATES FUSION

Each part	IP dataset	HU dataset	PU dataset
SAFD with BP	68.89	78.16	83.44
LBP	90.06	37.42	97.74
multi-scale Kalman filter	90.53	95.31	98.75
Different parts combinations	**IP dataset**	**HU dataset**	**PU dataset**
(SAFD with BP)+(multi-scale Kalman filter)	95.62	98.25	99.34
(SAFD with BP) +LBP	98.12	85.73	99.21
(SAFD without BP)+(multi-scale Kalman filter)+LBP	94.78	97.95	98.68
(SAFD with BP)+(multi-scale Kalman filter) +LBP	**98.93**	**98.88**	**99.93**

samples, ours achieves accuracies of 73.68% (IP), 65.59% (HU), and 82.52% (PU), substantially outperforming other methods (e.g., GSL: 63.62%, 33.68%, and 47.28%). As the sample size increases to 30, ours further improves to 96.64% (IP), 91.18% (HU), and 97.7% (PU), consistently maintaining its superiority. This indicates that the proposed method not only performs competitively with large training samples but also demonstrates robustness in data-scarce conditions through effective feature learning, highlighting its practical value for few-shot remote sensing image classification tasks.

2) Ablation Studies 1: The Impact of Each Part and Combinations of Different Parts on Classification Performance: We also analyzed the computational complexity of all comparison methods and the proposed method on the HU dataset, and the results are shown in Table IX. We can observe that the proposed method performs better than other comparative methods. From the perspective of balancing classification performance and computational efficiency, the proposed method achieves a significant advantage over all other compared methods while maintaining moderate training and testing time. Although its training time (346.41 s) is higher than lightweight traditional methods like GSL (35.28 s), it is far lower than complex deep models such as SAFDNet (2278.2 s) and DHTSRNet (1512.6 s). In testing, it requires only 2.03 s, second only to GSL (0.02 s) but significantly faster than other deep

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models (e.g., SAFDNet's 583.5 s). Crucially, the proposed method dominates all classification metrics (OA: 98.88%, AA: 99.06%, and κ: 98.79%), and its superior accuracy outweighs the minor increase in time cost. This equilibrium demonstrates that ours fulfills high-precision requirements without excessive computational overhead, making it a practical and competitive choice for real-world applications. As for parameter efficiency (Params): Our method (Ours) requires only 0.02 MB of parameters, which is significantly lower than most compared methods (e.g., 83.33 MB for SS_TMNet and 87.04 MB for Morphformer). This demonstrates our model's high memory efficiency and lightweight nature, making it highly suitable for deployment on devices with limited computational resources. In summary, while achieving the best classification performance, our method also exhibits superior parameter efficiency and excellent inference speed, demonstrating its outstanding computational efficiency advantage.

Feature combinations substantially enhance performance over individual methods. The fusion of SAFD with BP and a multiscale Kalman filter yields notable improvements (95.62% IP, 98.25% HU, and 99.34% PU), underscoring the complementary nature of depth features and multiscale information. Integrating LBP and SAFD with BP further boosts accuracy in IP (98.12%) and PU (99.21%), though HU performance remains moderate (85.73%) due to LBP's weakness in this dataset. Crucially, the exclusion of BP in SAFD degrades results (94.78% IP, 97.95% HU, and 98.68% PU), confirming BP's essential role in optimizing depth features. The full combination of all three components—SAFD with BP, multiscale Kalman filter, and LBP, which achieves peak performance across all datasets (98.93% IP, 98.88% HU, and 99.93% PU), demonstrating that their synergistic fusion maximizes classification accuracy by effectively integrating depth, multiscale spatial, and texture features. This comprehensive approach consistently outperforms all other configurations, validating the necessity of combining these complementary feature types for robust HSI classification.

3) Ablation Studies 2: The Impact of SAFD With and Without BP: To investigate the impact of each part and combinations of different parts in the proposed method on classification accuracy, we conducted ablation studies on three datasets: IP, HU, and PU. The results are presented in Table X. Based on the experimental results presented in the table, the analysis of individual parts and their combinations reveals several key observations regarding their impact on HSI classification performance. The multiscale Kalman filter consistently demonstrates strong individual performance across all datasets, achieving 90.53% (IP), 95.31% (HU), and 98.75% (PU), indicating its fundamental importance for capturing multiscale information. In contrast, SAFD with BP shows limited standalone effectiveness with the lowest individual scores (68.89% IP, 78.16% HU, and 83.44% PU), while LBP exhibits significant dataset—dependent variability, excelling in PU (97.74%) and IP (90.06%) but performing poorly in HU (37.42%).

4) Computational Efficiency: We also verified the results of SAFD with and without BP in the experiment. The experimental result is shown in Rows 8–9 of Table X. Based on the comparative results presented in the table, the integration of the BP algorithm with the SAFD method demonstrably enhances model performance across all three datasets. The SAFD model with BP consistently achieves higher accuracy, outperforming its non-BP counterpart by a significant margin in the IP dataset (increasing from 94.78% to 98.93%) and securing superior results in the HU and PU datasets (98.88% and 99.93%, respectively). This performance gain provides compelling evidence that enabling learnable parameters within the SAFD framework through BP allows the model to adaptively refine its feature extraction process, leading to a more powerful and discriminative representation of the data. The results confirm that the synergistic combination of SAFD and BP is crucial for unlocking the model's full potential and achieving state-of-the-art classification accuracy.

TABLE XI
COMPARISON OF COMPUTATIONAL TIME AND OA ACROSS DIFFERENT DATASETS

Experiments with same parameters			
	IP	PU	HU
Time(s)	3.87	26.26	97.69
OA(%)	99.05	99.92	98.50
Experiments with different parameters			
	IP	PU	HU
Time(s)	3.20	26.26	118.27
OA(%)	98.93	99.93	98.88

5) Analysis of the Superior Performance: The superior performance of our method can be attributed to its effective fusion strategy that harmoniously integrates features from SAFD with BP, multiscale Kalman filtering, and LBP. This is quantitatively substantiated by the ablation study in Table X, which demonstrates that the combination of all three components yields the highest accuracy, indicating strong synergistic effects. Specifically, the multiscale Kalman filter provides robust denoising and enhances spatial coherence, which explains the significant reduction in noisy misclassifications visible in our result maps compared to others. Furthermore, the computational efficiency presented in Table IX shows that our method achieves a favorable balance, offering top-tier accuracy with competitive training and, notably, very low inference times, second only to the simplistic GSL method. This efficiency stems from the design of our feature fusion network, which avoids the complexity of very deep architectures or expensive transformer-based attention mechanisms used in methods like ACTN or Morphformer, while still effectively capturing both spectral details and spatial context. The instability of some comparator methods (e.g., Morphformer's performance drop on HU) may arise from their sensitivity to data distribution or hyperparameters, whereas our handcrafted-feature-inspired yet learned fusion approach exhibits greater robustness across diverse scenes.

6) Analysis of Scalability Experiments: To comprehensively address this concern, we conducted dedicated experiments evaluating the model's adaptability to large-scale hyperspectral data and diverse parameter configurations. As presented in Table XI, we tested three datasets with drastically

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TABLE XII

IMPACT OF DIFFERENT TEXTURE FEATURE EXTRACTION METHODS ON CLASSIFICATION PERFORMANCE (OA%)

Methods	IP	HU	PU
GLCM	98.33	97.53	99.80
Tamura	97.53	97.39	99.79
Wavelet	97.82	97.44	99.78
Laws	98.13	97.11	99.81
LBP	**98.93**	**98.88**	**99.93**

TABLE XIII

INFLUENCE OF DIFFERENT NOISE LEVELS ON CLASSIFICATION PERFORMANCE ON IP DATASET

Method	no noise	5dB	10dB	15dB	20dB
Traditional SAFD	95.49	93.84	93.70	93.91	94.58
Ours method	98.93	98.85	98.76	98.84	98.87

increasing spatial dimensions: IP (145 × 145, 21K pixels), PU (610 × 340, 207K pixels), and HU (349 × 1905, 665K pixels). The results reveal critical insights into scalability. Under fixed parameters (four layers, three decompositions, 3 × 3 patches), computational time scales sublinearly with input size: while HU's spatial volume is 31.7× larger than IP, its processing time increases only 25.2× (3.87 ⟶ 97.69 s). This demonstrates the model's efficient handling of exponentially growing data. Furthermore, when optimizing parameters per dataset (e.g., seven layers for HU), the model achieves a higher accuracy of 98.88% (+0.38% OA) with a marginal 21.1% time increase (97.69 ⟶ 118.27 s), highlighting its ability to balance precision and efficiency for massive inputs. Crucially, OA remains consistently high (>98.50%) across all scales and configurations, proving robustness without compromising performance. These findings confirm our method's practical scalability for real-world large-scale hyperspectral applications, efficiently processing datasets exceeding 660K pixels while maintaining competitive accuracy through adaptable parameterization.

7) Comparison Among Different Texture Feature Extraction Methods: To further validate LBP's effectiveness, we conducted comprehensive comparative experiments evaluating LBP against four prominent alternative texture feature extraction methods: gray-level co-occurrence matrix (GLCM), Tamura texture features, wavelet transform, and laws' texture energy measures. The classification performance is summarized in Table XII. The results consistently demonstrate that our implemented LBP method outperforms other texture feature extraction methods across all three HSIs datasets. Specifically, LBP achieved the highest OA% values (98.93% for IP, 98.88% for HU, and 99.93% for PU), exceeding the performance of GLCM, Tamura, wavelet, and laws features on these datasets. This comparative analysis strongly supports the effectiveness and robustness of the LBP approach for texture feature extraction within the context of our methodology.

8) Robustness of Features: Table XIII explicitly compares the classification performance degradation under varying levels of additive noise. Traditional SAFD shows a clear sensitivity: performance drops from 95.49% (no noise) to 93.70% at 10-dB noise (a drop of 1.79%), and only recovers partially to 94.58% at 20 dB. This fluctuation of 1.79% demonstrates its vulnerability to noise corruption. In stark contrast, our method exhibits exceptional stability: performance remains remarkably consistent, starting at 98.93% (no noise) and experiencing only a minuscule degradation to 98.76% at 10 dB (a drop of just 0.17%) and 98.87% at 20 dB. The maximum observed degradation is merely 0.17%, which is over an order of magnitude smaller than the degradation seen with traditional SAFD. This minimal fluctuation under significant noise levels (5–20 dB) provides compelling empirical evidence for the superior noise robustness achieved by our BP-optimized kernel, aligning perfectly with our theoretical argument.

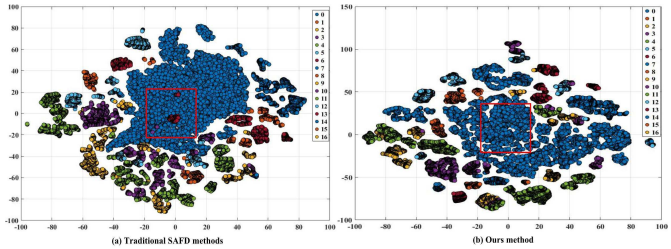


Fig. 11. t-SNE visualization of feature space on IP dataset.

9) Feature Discriminative Power: We employed t-SNE visualizations to qualitatively assess the separability of the features extracted by both methods. The t-SNE plots in Fig. 11 reveal a pronounced difference in discriminative capability. Features extracted by the traditional SAFD method show noticeable overlap and less distinct cluster boundaries, particularly for the challenging category highlighted by the red rectangle, which is poorly separated from neighboring classes. Conversely, the features extracted by our method demonstrate significantly enhanced cluster compactness and separation. Individual classes form tight, well-defined clusters with substantially larger interclass margins. Crucially, the category within the red rectangle, which was ambiguous using traditional SAFD features, is now clearly and distinctly separated from others in our feature space. This visual evidence strongly supports our claim that the BP-optimized kernel in our method extracts features with superior discriminative power—maximizing intraclass similarity (small intraclass distances) while maximizing interclass dissimilarity (large interclass distances), leading to features that are inherently easier to classify.

10) Influence of the Proportion of Training Samples on Classification Performance: To further demonstrate the adaptability of our proposed method across varying the proportion of training samples (ranging from small-scale to medium-scale), this article additionally includes an analysis of classification performance under different proportions of training samples, and the results are shown in Fig. 12. Experimental results demonstrate that as the training sample ratio increases from 0.1% to 7%, all methods show improved classification performance, though the magnitude of improvement and final performance vary significantly across datasets and approaches. On the most challenging HU dataset, methods generally exhibit low initial

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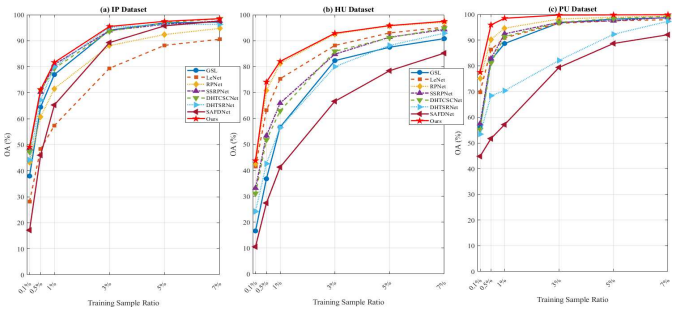


Fig. 12. Influence of different proportions of training samples on classification performance on three datasets.

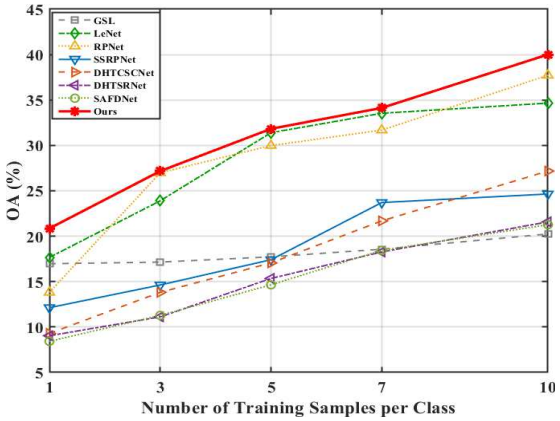


Fig. 13. Few-shot performance on HoustonU2018 dataset.

performance at the minimal 0.1% ratio (with the lowest at just 10.52%), yet the proposed method demonstrates exceptional adaptability—rising from 43.74% to 97.47%. Notably, at the 0.5% ratio, it achieves 74.06%, substantially outperforming alternatives (the next highest being RPNet at 70.73%). For the IP dataset, traditional methods like GSL and LeNet show weaker low-sample performance (37.99% and 28.23% at 0.1%, respectively), while the proposed method consistently maintains leadership, ultimately reaching 98.50% accuracy to surpass DHTCSCNet's 98.42%. All methods perform well on the PU dataset, but the proposed method exhibits remarkable dominance: achieving near-perfect 99.87% accuracy at 7% samples, with an 18.4% performance leap (77.49% $\longrightarrow$ 95.89%) between 0.1% and 0.5% ratios—indicating highly efficient feature learning under extreme sample scarcity. Interestingly, DHTSRNet shows anomalous fluctuation on PU data, increasing from 68.38% to 70.33% when samples increase from 0.5% to 1%, while SAFDNet consistently underperforms on HU (peaking at only 85.19%), revealing certain methods' sensitivity to specific data distributions. Collectively, the proposed method maintains optimal performance across all sample ratios and datasets, demonstrating an average advantage of 7.82% over suboptimal methods in the resource-constrained 0.1% scenario—highlighting exceptional few-shot learning capability and generalization robustness.

11) Generalization Ability: The visualization of Fig. 13 demonstrates that the proposed method consistently outperforms all comparison methods across all few-shot settings

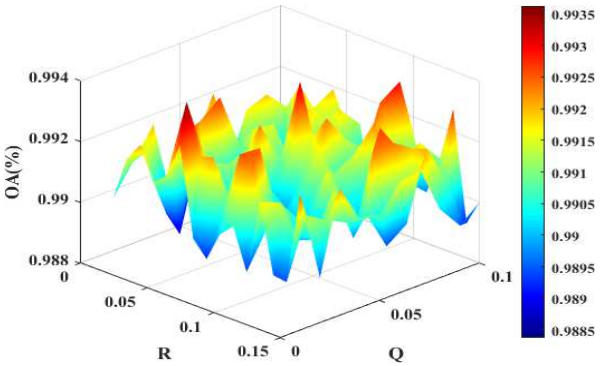


Fig. 14. Effect of hyperparameters $Q(0.01 - 0.10)$ and $R(0.01 - 0.15)$ combinations on classification accuracy on IP dataset.

(1–10 samples per class) on the challenging HoustonU2018 dataset. Notably, our approach achieves a substantial performance margin of 3.19~16.44 percentage points at the extremely challenging 1-sample setting (20.86% versus 8.42%~17.67% for other methods), highlighting its superior feature representation capability under severe data scarcity. As training samples increase to 10 per class, our method maintains its leading position with a significant 2.29~5.29 percentage point advantage (40.01% versus 20.25%~37.72% for others), confirming its robustness in complex, large-scale scenarios. The consistent performance gap across all sample counts suggests that our architecture effectively addresses key challenges in few-shot remote sensing classification, including limited supervision, high intra-class variability, and complex spatial–spectral relationships inherent in large-scale hyperspectral datasets. This advantage is particularly remarkable when considering that HoustonU2018 represents a more complex and realistic testing environment compared to standard benchmark datasets. This also verifies that the proposed method has good generalization performance.

12) Sensitivity Analysis of Hyperparameters Q and R Combinations on Classification Accuracy: The settings of Q and R are actually the settings of hyperparameters. Next, we will verify their impact on classification performance through experiments, and the experimental results are shown in Fig. 14. The experimental results demonstrate significant sensitivity of classification accuracy (OA%) to hyperparameters $Q(0.01-0.10)$ and $R(0.01-0.15)$, with performance variations exceeding 0.5% across the parameter space. Peak accuracy (99.36%) occurs at $Q = 0.02/R = 0.05$, while the lowest performance (98.87%) appears at both $Q = 0.10/R = 0.07$ and $Q = 0.03/R = 0.03$. Lower Q values $(0.01 - 0.03)$ exhibit greater stability across R variations, particularly $Q = 0.02$, which maintains $> 99.1\%$ accuracy for nearly all R values. Higher Q settings $(0.08 - 0.10)$ show increased volatility, especially with lower R values, where performance can drop below 99%. Mid-range R values $(0.04 - 0.08)$ generally yield optimal results, with $R = 0.05$ emerging as particularly robust. The interaction pattern reveals a diagonal performance ridge from $(Q = 0.02/R = 0.05)$ to $(Q = 0.07/R = 0.15)$, suggesting compensatory effects between parameters. Isolated performance spikes occur at unexpected locations like

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$Q = 0.07/R = 0.15$ (99.23%), while performance valleys concentrate in low-Q/low-R and high-Q/low-R regions. These nonlinear interactions indicate that combinatorial optimization is essential, with the optimal zone centered at $Q = 0.02 - 0.03$ and $R = 0.05 - 0.08$, where accuracy consistently exceeds 99.2%. The substantial performance variations highlight that hyperparameter tuning can critically impact classification effectiveness in precision-sensitive applications.

V. Conclusion

In this article, we propose a deep SAFD network with BP, where the network parameters are automatically adjusted in an unsupervised manner using the SAFD with BP algorithm. This approach resolves the manual parameter tuning problem in conventional pretraining methods, and concurrently reduces the dependence on extensive labeled datasets that modern end-to-end deep learning frameworks typically require for network parameter optimization. Furthermore, we develop a 3-D Kalman filter pyramid for hyperspectral feature extraction and rotation-invariant LBP texture enhancement, then integrate these with deep SAFD-BP network features into an optimized classification framework. Experimental results show that the proposed method achieves higher classification accuracy among all compared methods.

In the future, we will focus on lightweight deployment, the development of a unified SAFD-based deep network framework that explicitly extracts and deep spatial–spectral features.

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